UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 6: Linear Time Invariant Systems

Reference: Chapter 2
(Section 2.4 – 2.5)

Please turn off your gadgets now

Recap: Impulse Response of LTI System

The convolution integral is a direct formula for calculating the response of an LTI system to *any* input knowing its impulse response $h(t)$

$x(t)$ **Input** → **LTI System** $h(t)$ → **Output** $y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$

$$= x(t) * h(t)$$

Remember:

1. **It only works for Linear Time-invariant (LTI) systems, since we needed all 3 LTI properties to derive this formula**
2. **The formula assumes zero initial conditions**

Today: Graphical Convolution

The convolution integral has a geometric interpretation:

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\,d\tau$$

$$= \int_{-\infty}^{+\infty} x(\tau)h(-(\tau - t))\,d\tau$$

TIME REVERSAL

MULTIPLY

DELAY BY *t*

SUMMATION

Why Bother with the Graphical Method?

- Sometimes, the impulse response is available as a chart from an experiment rather than a formula
  - Easy to digitize into a numerical form
- The graphical method provides another way to understand convolution
  - Good for building intuition, and for checking our answers
- Graphical convolution provides a way to figure out or sanity-check the limits of convolution integrals

Graphical Convolution Steps

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

Keep in mind that:

- We are integrating the product of $x(\tau)$ with a flipped and shifted version of $h(\tau)$
- The integration is with respect to τ (t fixed), and the result y(t) is a function of t

Graphical convolution steps:

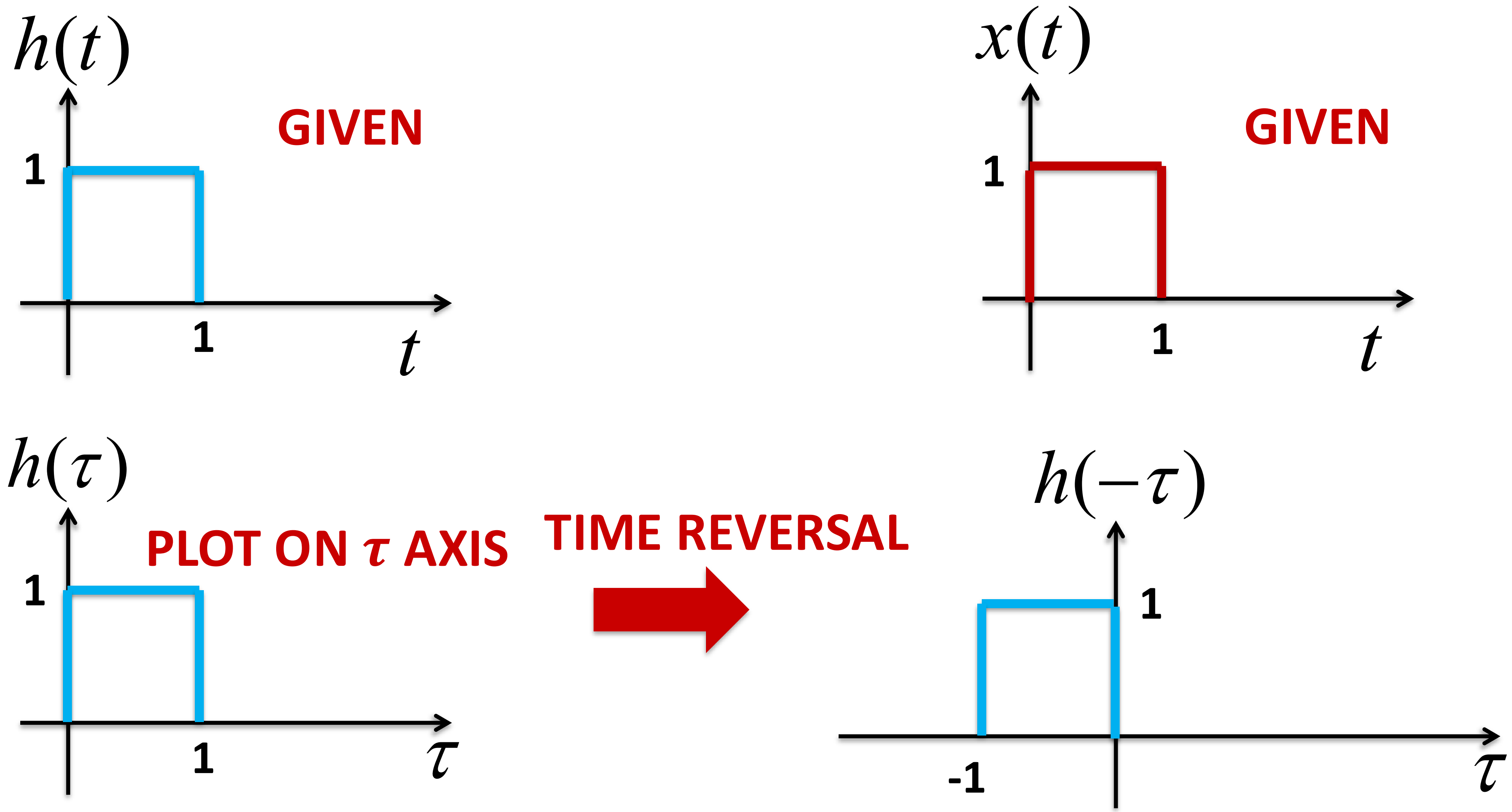
Step 1: On the τ axis, plot $x(\tau)$ and $h(-\tau)$ (flipped version of impulse response)

Step 2: Shift $h(-\tau)$ to the right by t to obtain $h\big(-(\tau - t)\big) = h(t-\tau)$

Step 3: Multiply $x(\tau)$ by $h(t-\tau)$ from Step 2, and integrate the product over τ

Step 4: Repeat the above for all possible values of t to obtain $y(t)$

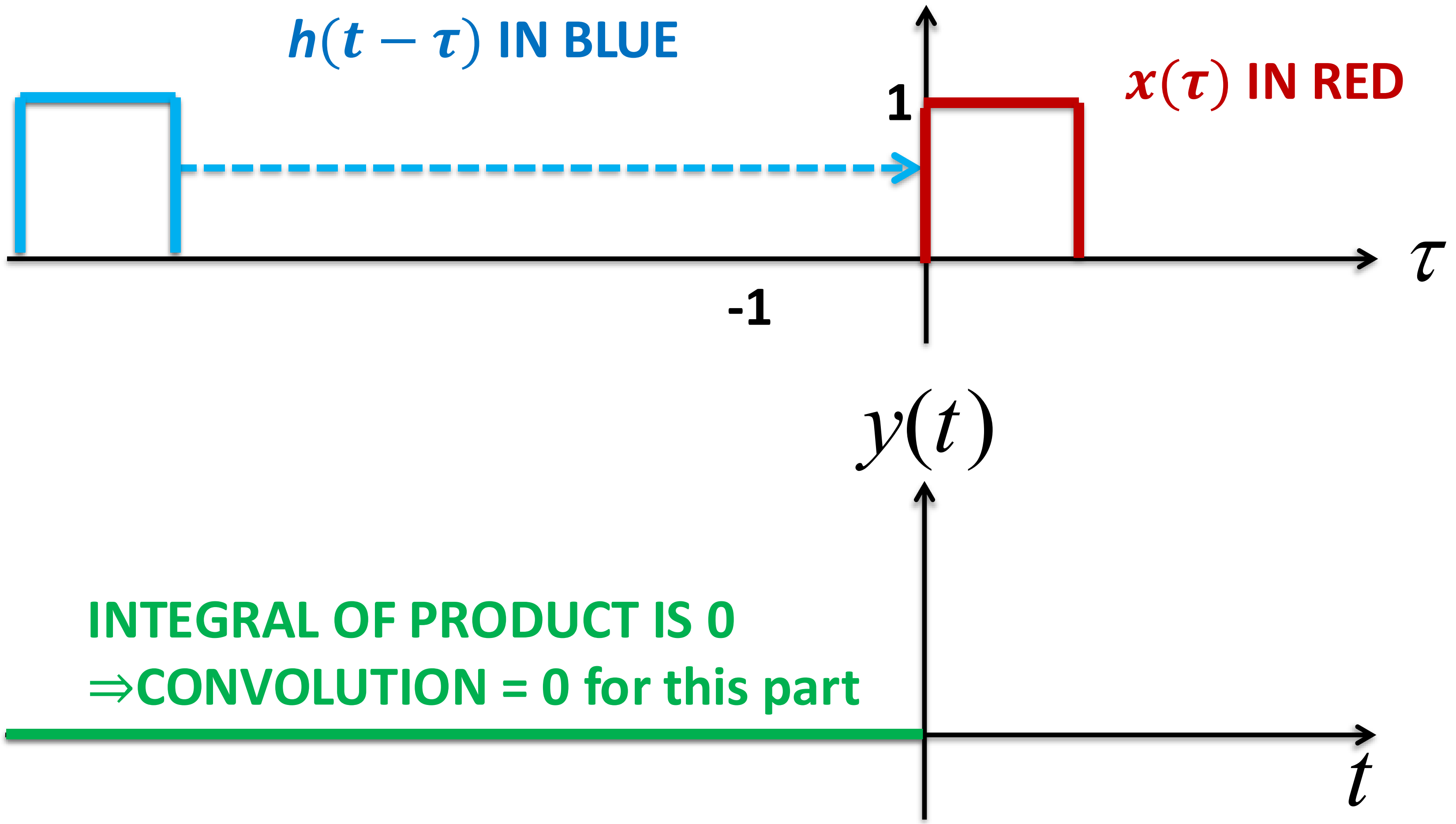
Graphical Convolution Example



Sliding: Part 1

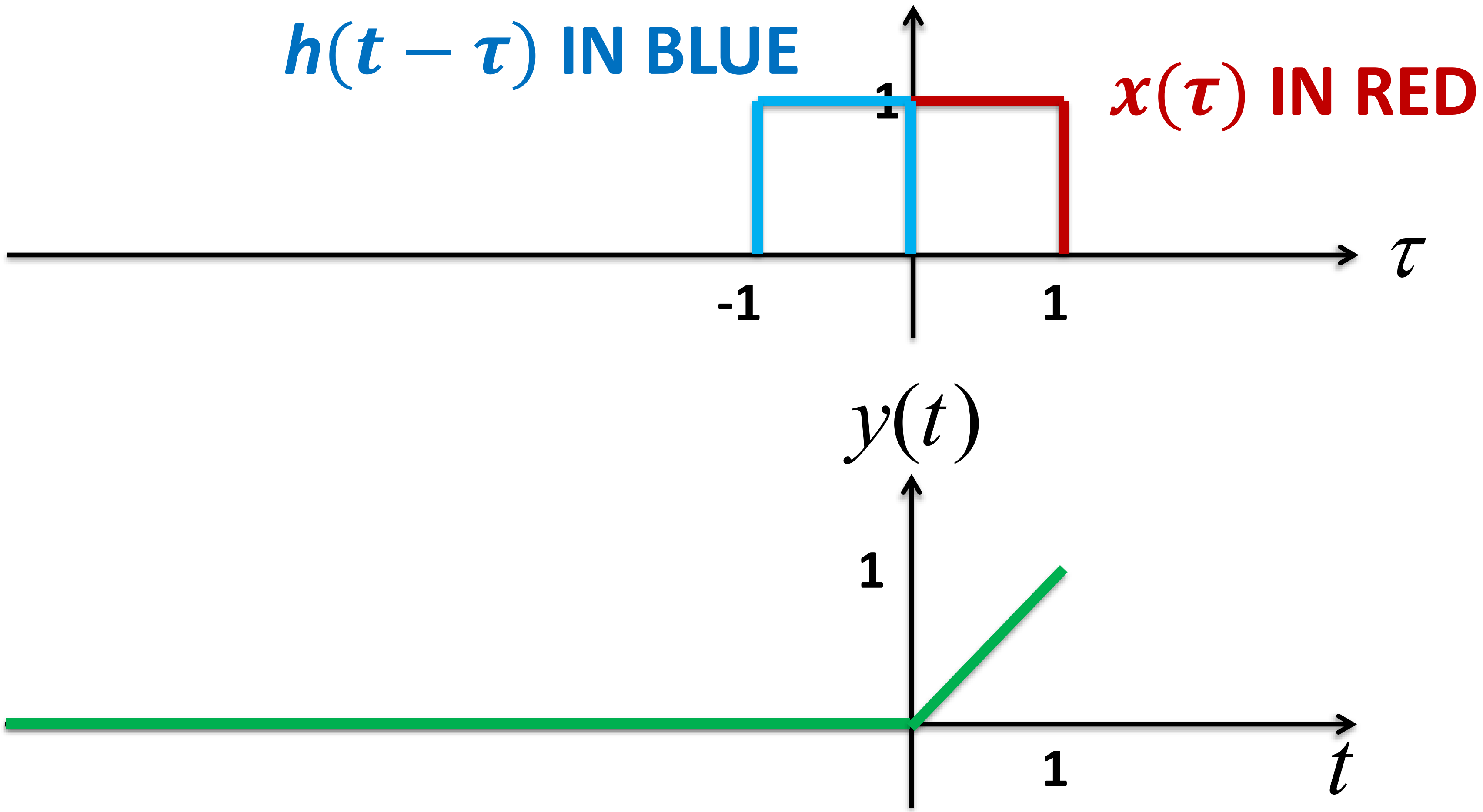
Slide the flipped impulse response from $t = -\infty$ to $t = 0$

During this part of the sliding, there is no overlap, so the product is zero



Sliding: Part 2

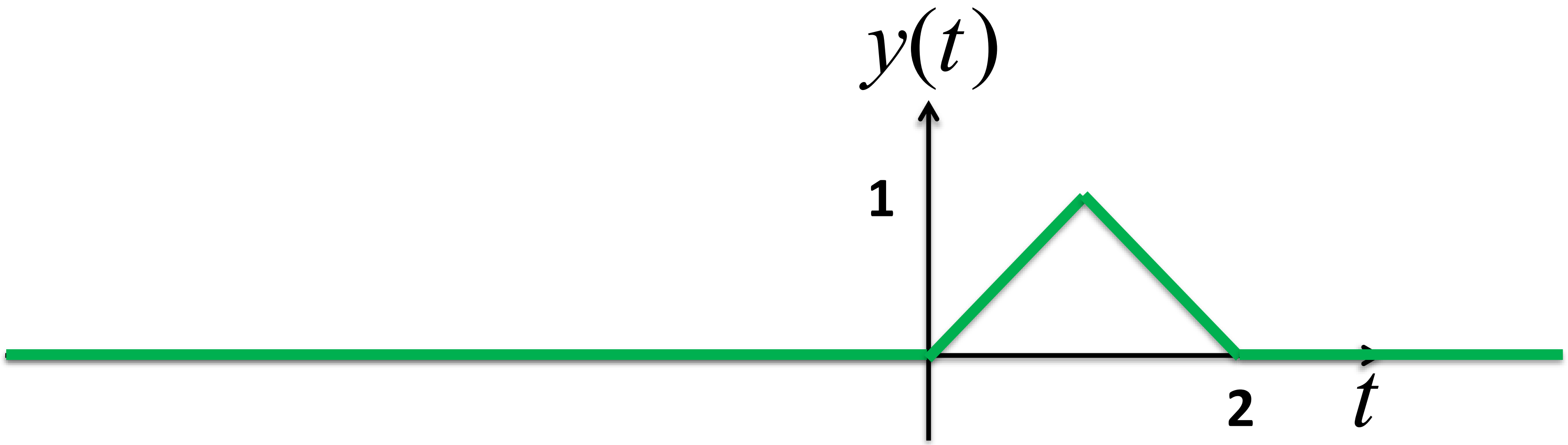
Between $t = 0$ and $t = 1$, the signal and impulse response intersect and the intersection duration is t until it reaches a maximum at $t = 1$



Sliding: Part 3

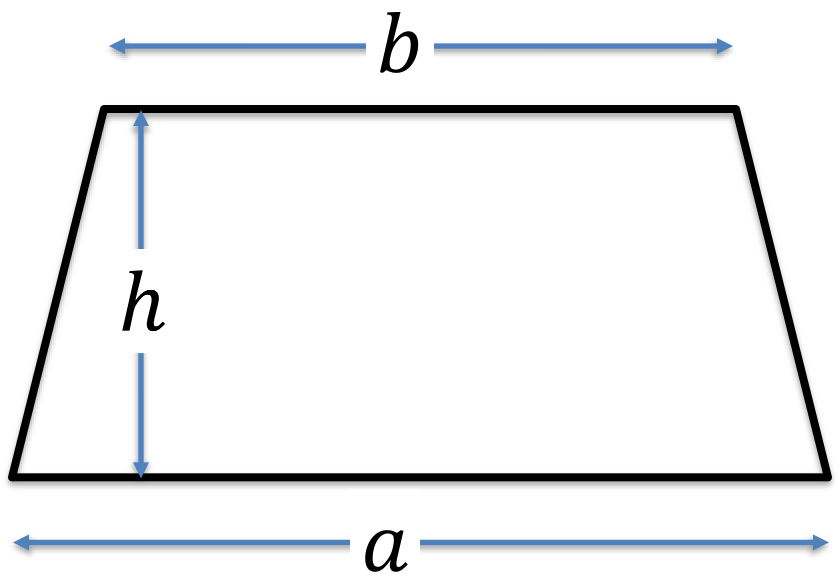
Between $t = 1$ and $t = 2$, the overlap between $x(\tau)$ and $h(t-\tau)$ decreases and reaches 0 at $t = 2$

For $t > 2$, there is no more overlap



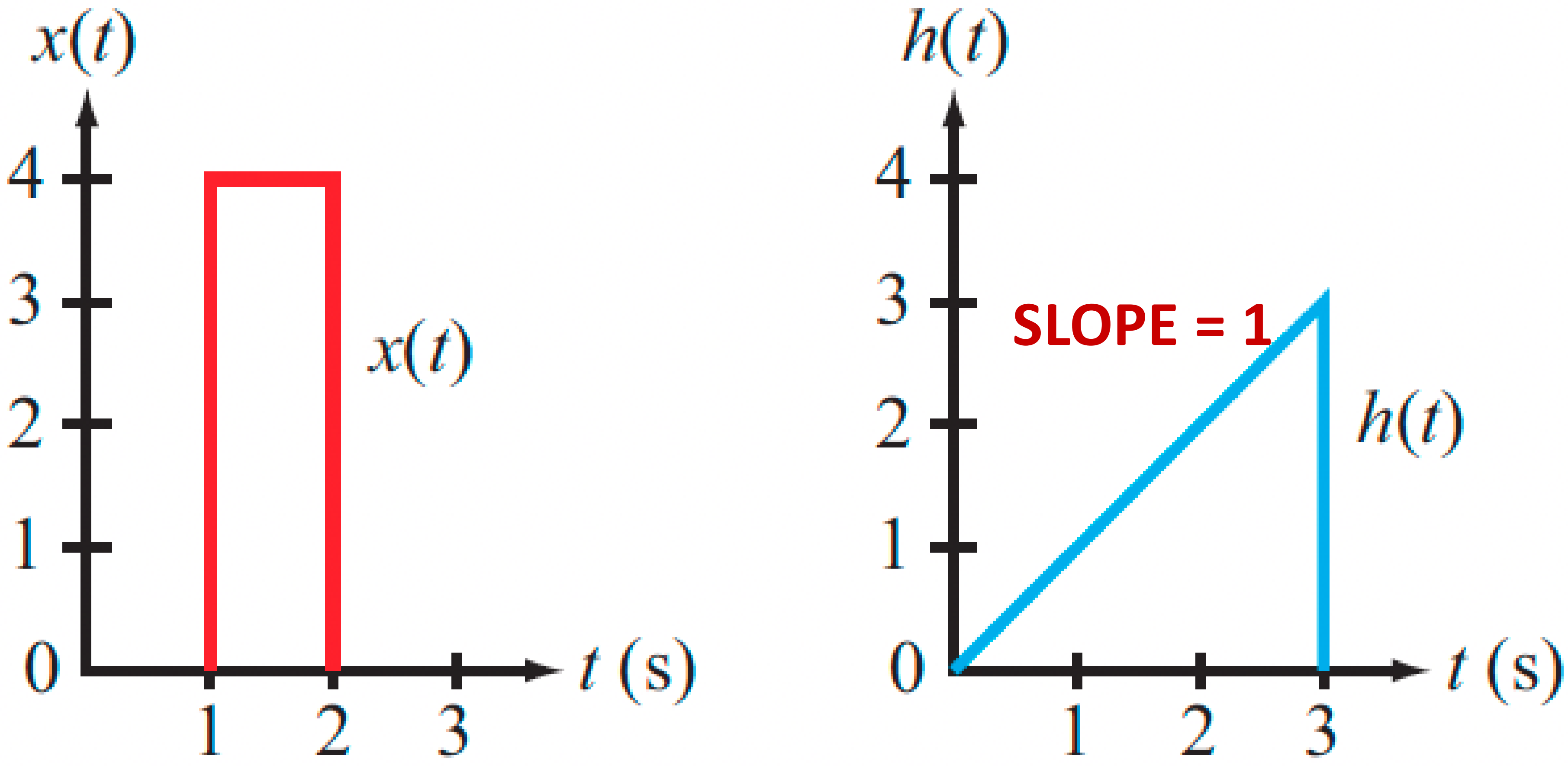
Math Moment

The area of a trapezoid



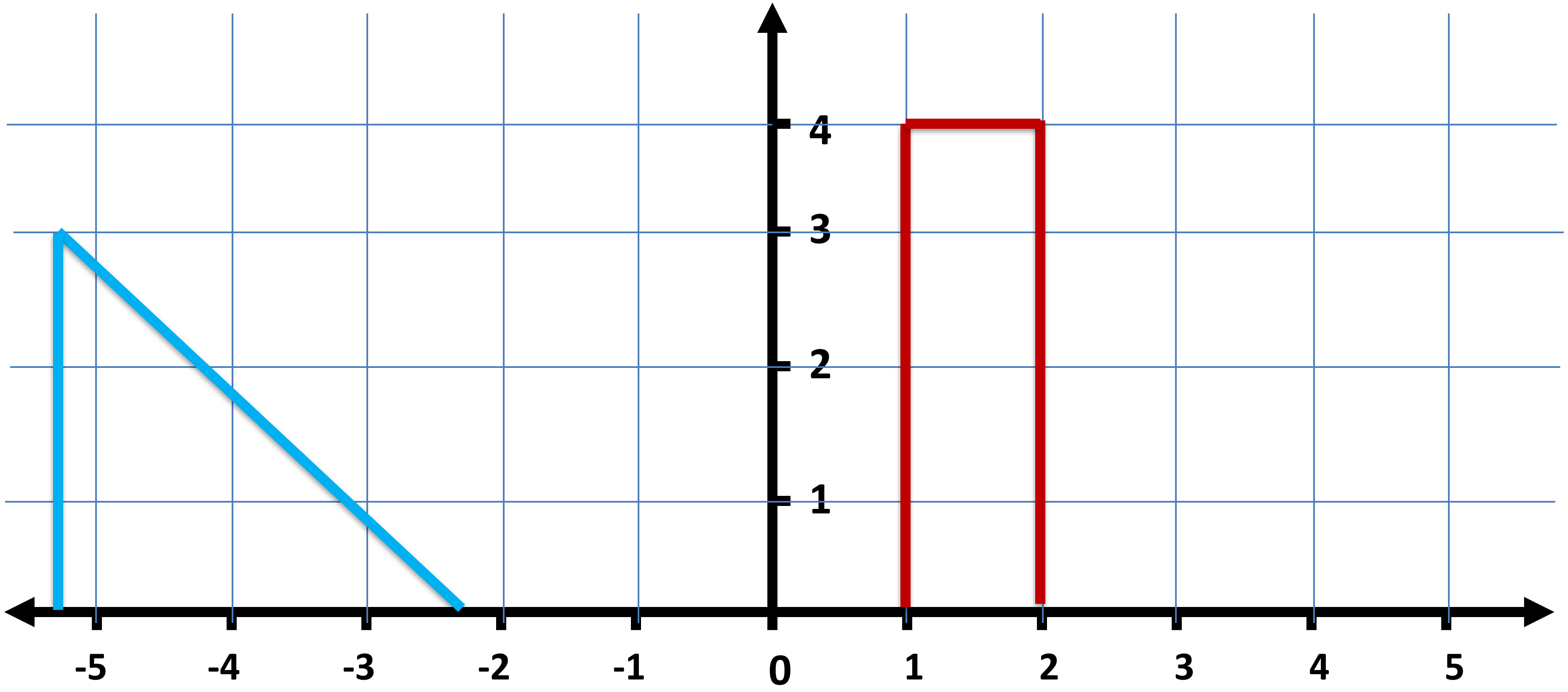
$$area = \frac{1}{2}h(a+b)$$

Another Example

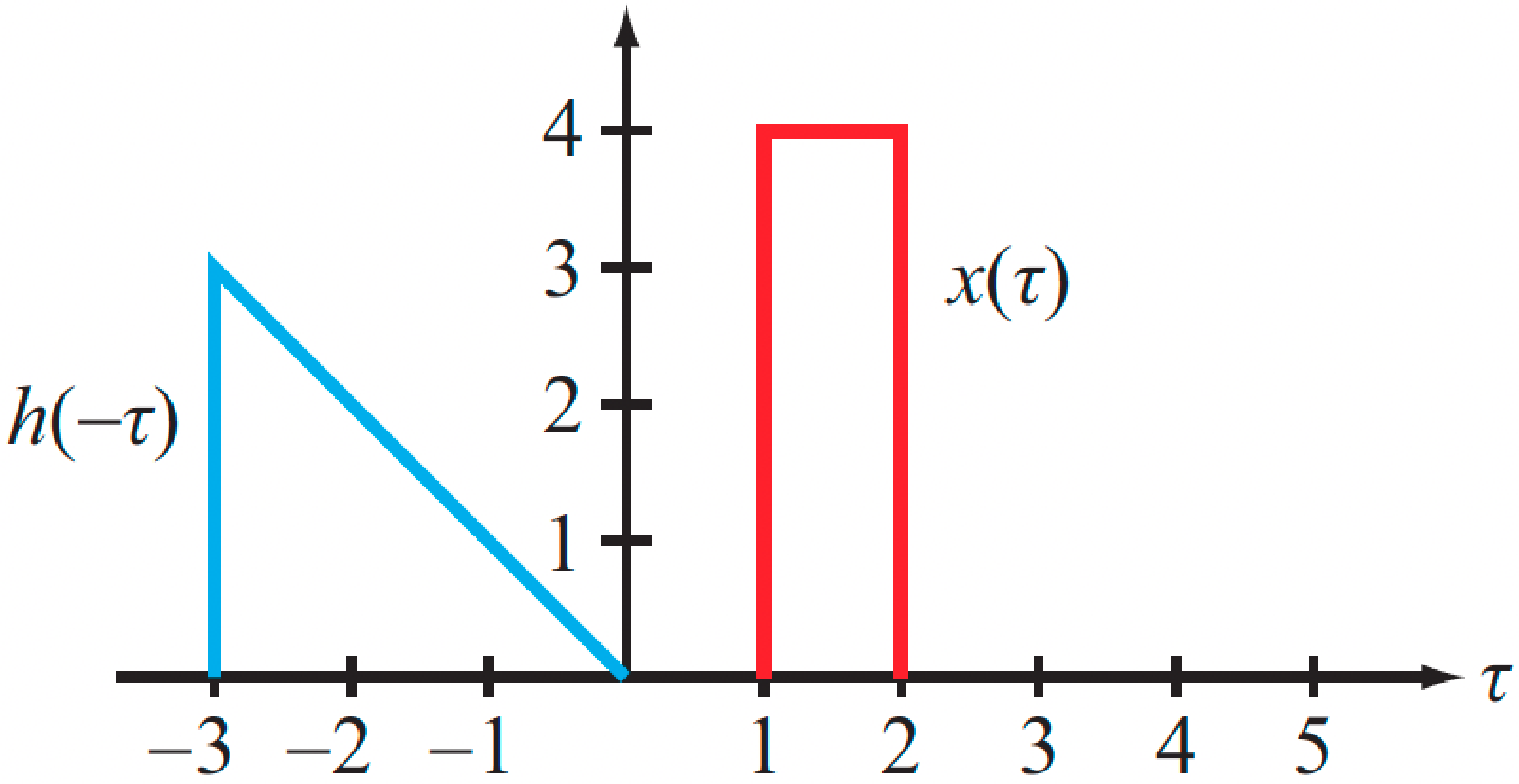


Determine $y(t) = x(t) * h(t)$

Animation



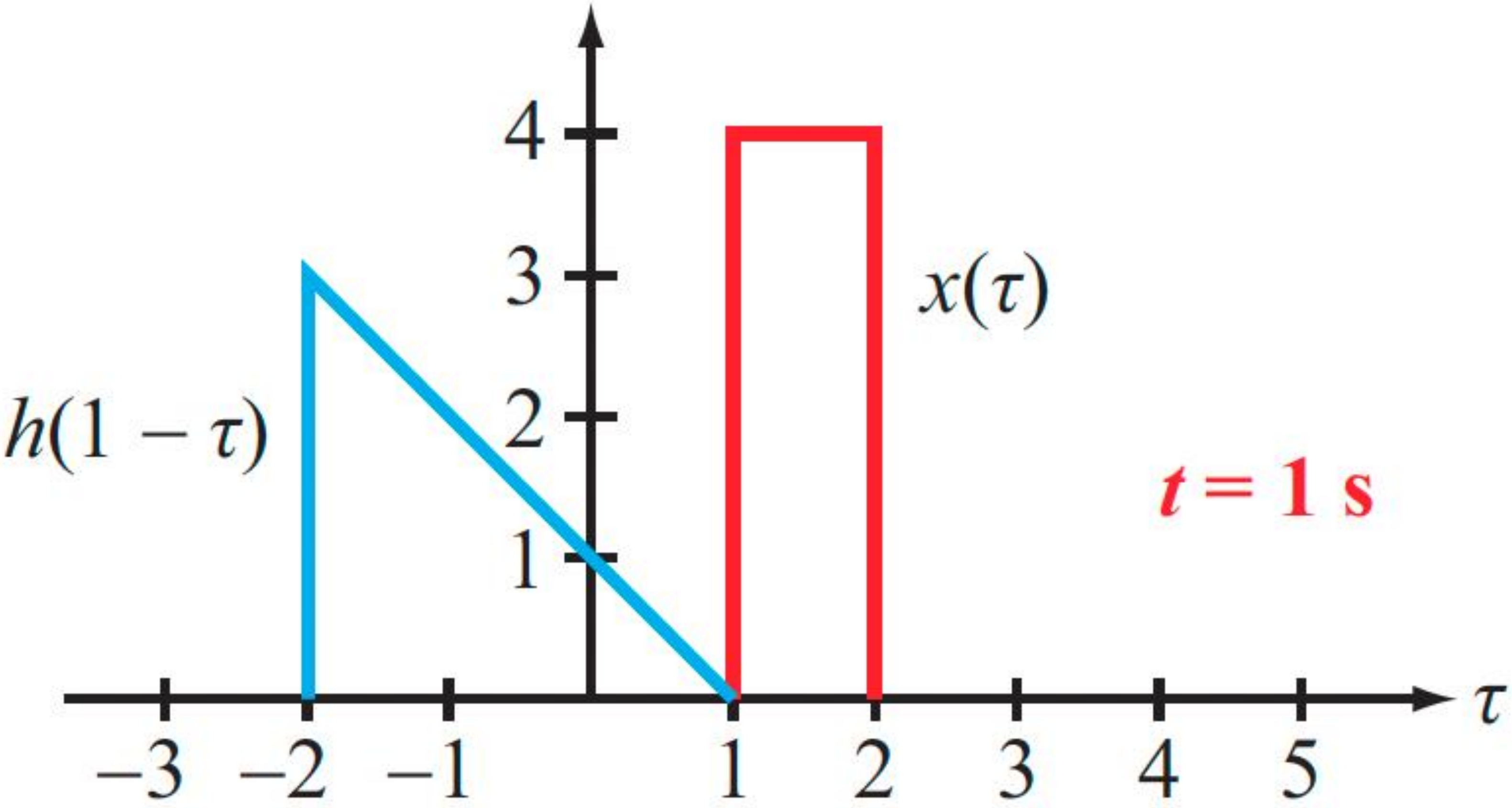
Flip the Impulse Response & Slide



$y(t) = 0$ for $-\infty < t < 0$

$x(\tau)h(t-\tau) = 0$ for these t values (no overlap)

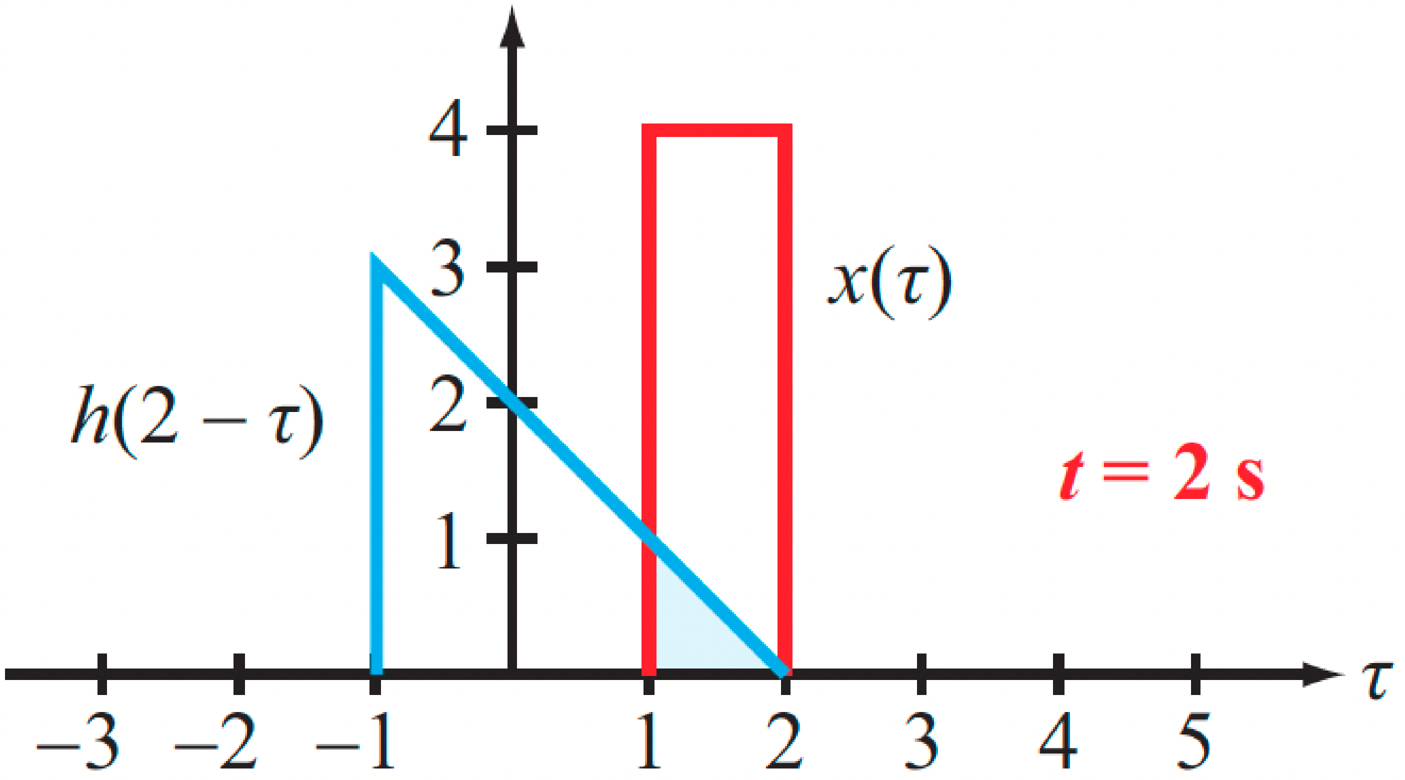
Flip the Impulse Response & Slide



$$y(t) = 0 \text{ for } -\infty < t < 1$$

$x(\tau)h(t-\tau) = 0$ for these t values (no overlap)

From t = 1 to t = 2



For $1 < t < 2$, the overlap is $(t-1)$ sec

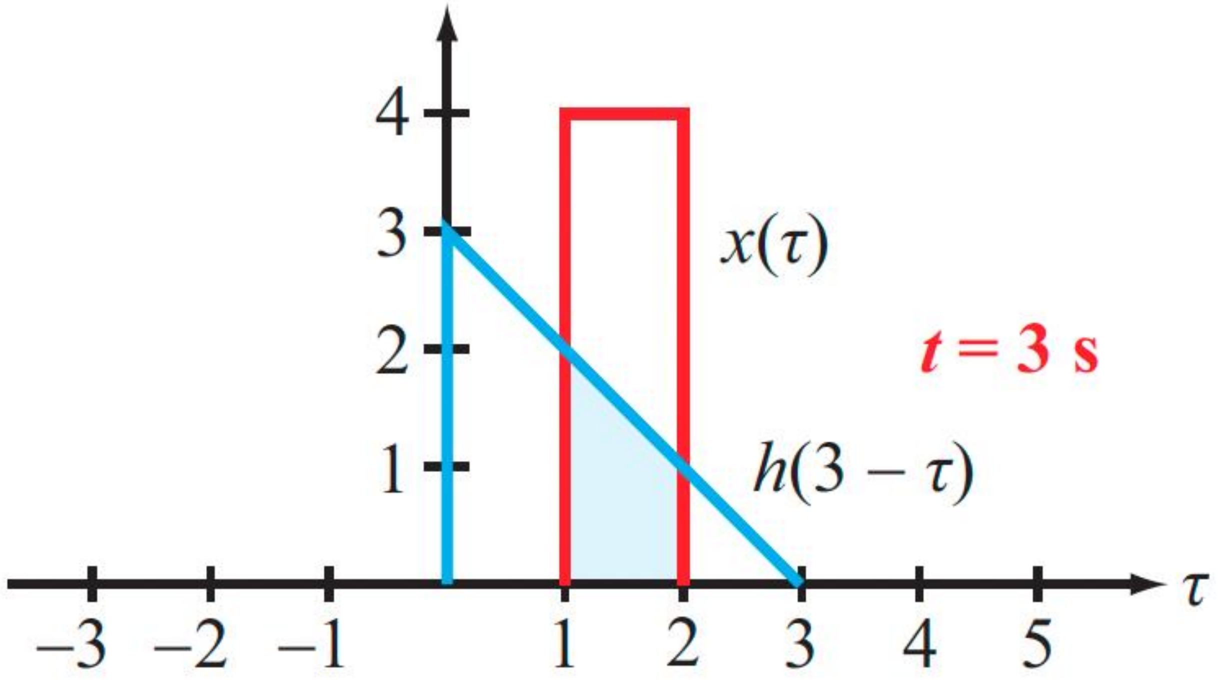
$y(t) = \frac{1}{2}(t-1)^2 \cdot 4$, with a maximum at $t = 2$ of 2

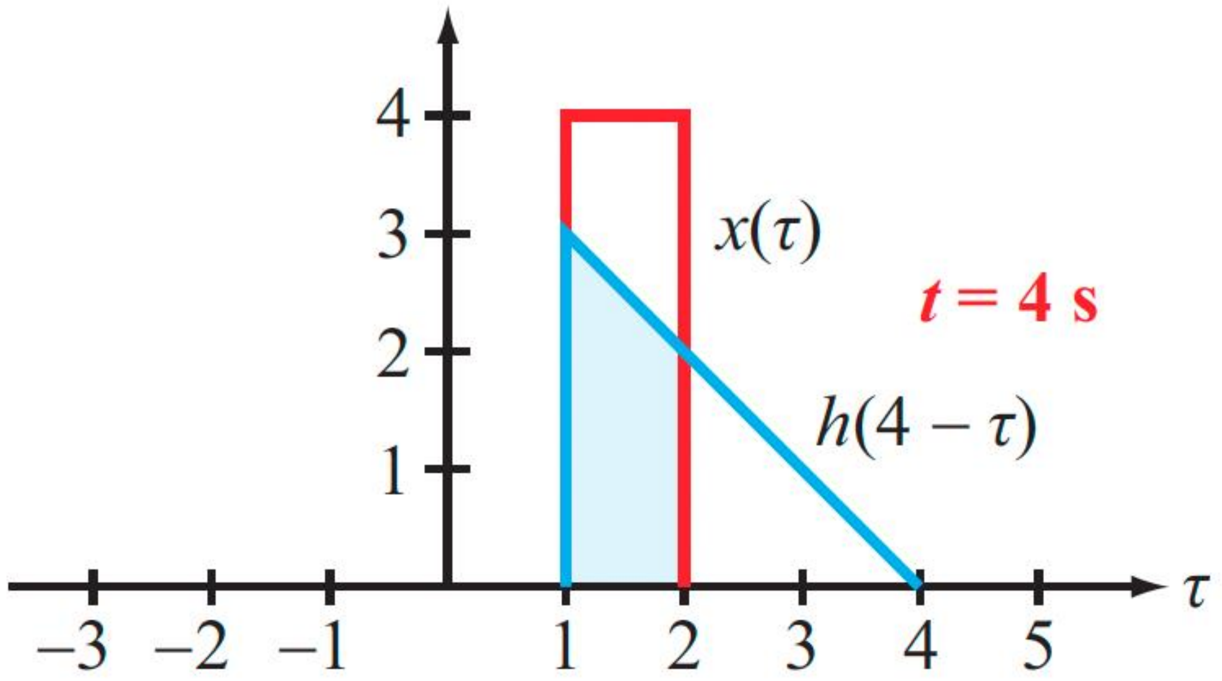
CAUTION

Keep in mind that we are <u>NOT</u> calculating the "area of the overlapping region", but rather, we are

"integrating the product of the two waveforms over the overlapping region"

From t = 3 to t = 4

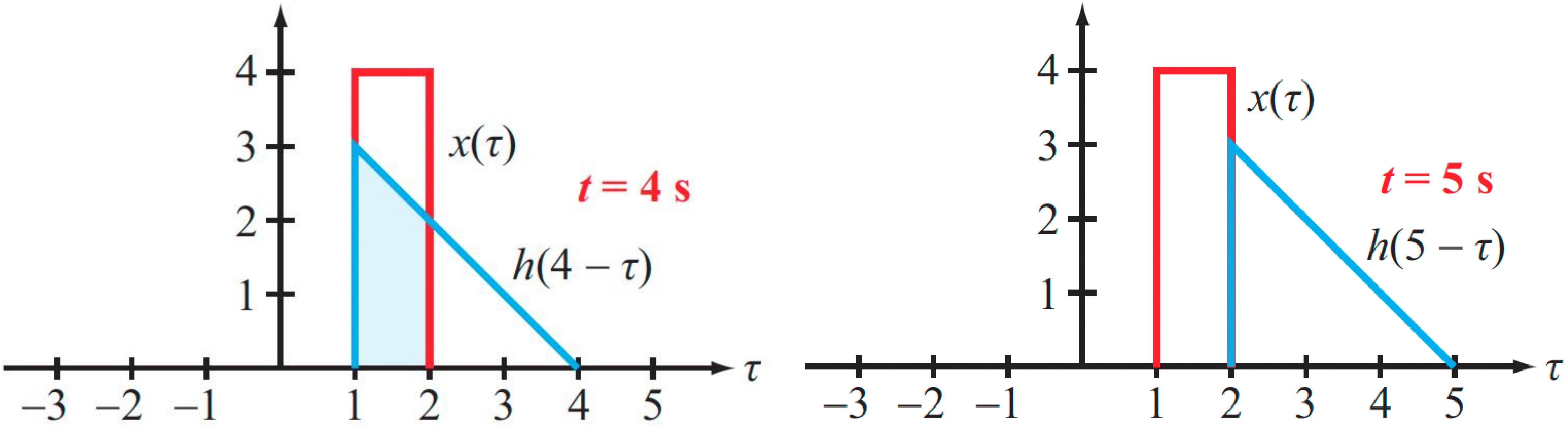




For $3 < t < 4$, the overlap is a trapezoid

$y(t)$ goes linearly from 2 to $\frac{(2+3)}{2} \cdot 1 \cdot 4 = 10$

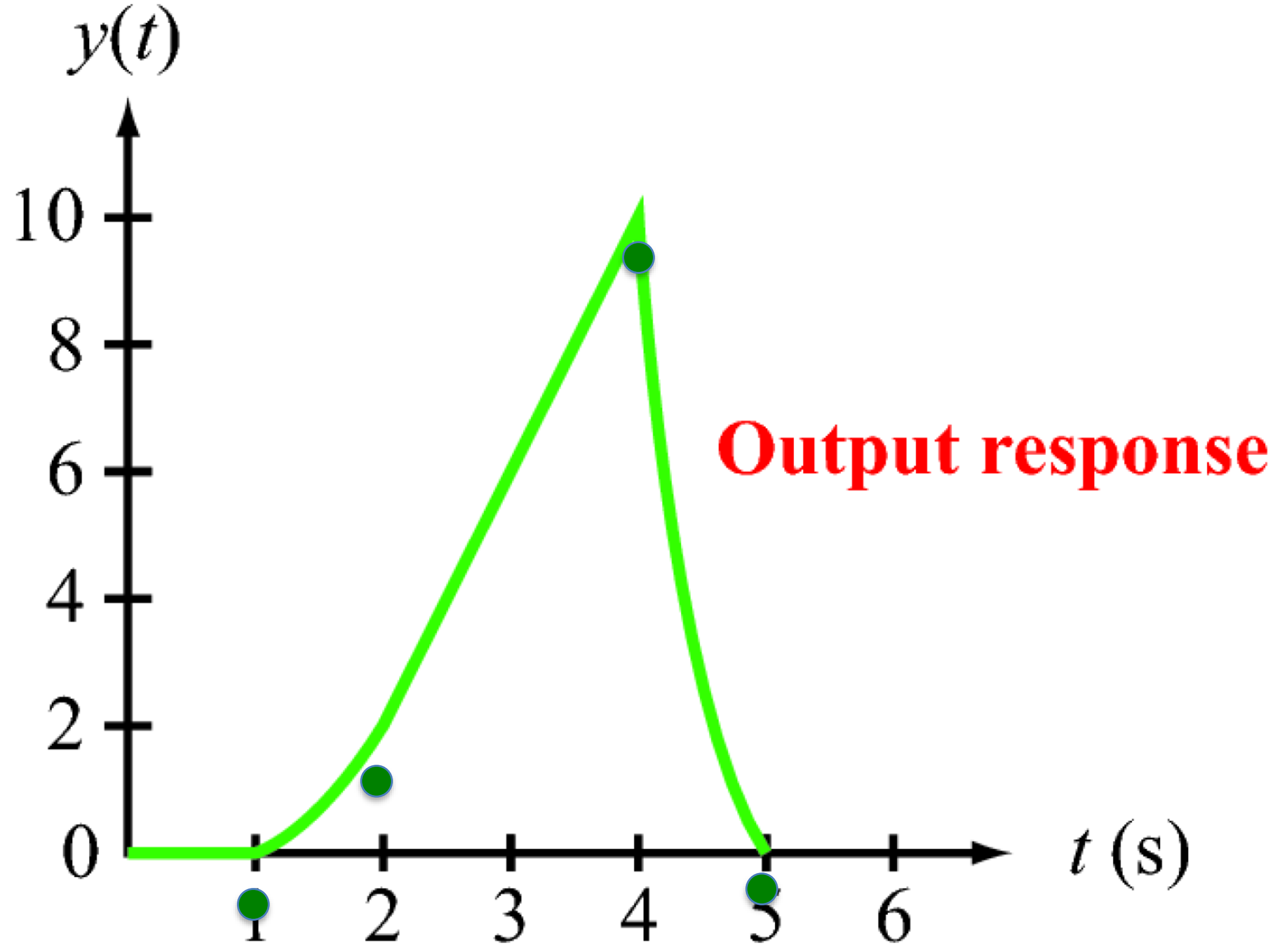
From t = 4 to t = 5



From $4 < t < 5$, there is an overlap

$y(t)$ falls quadratically from 10 to 0

Overall Response

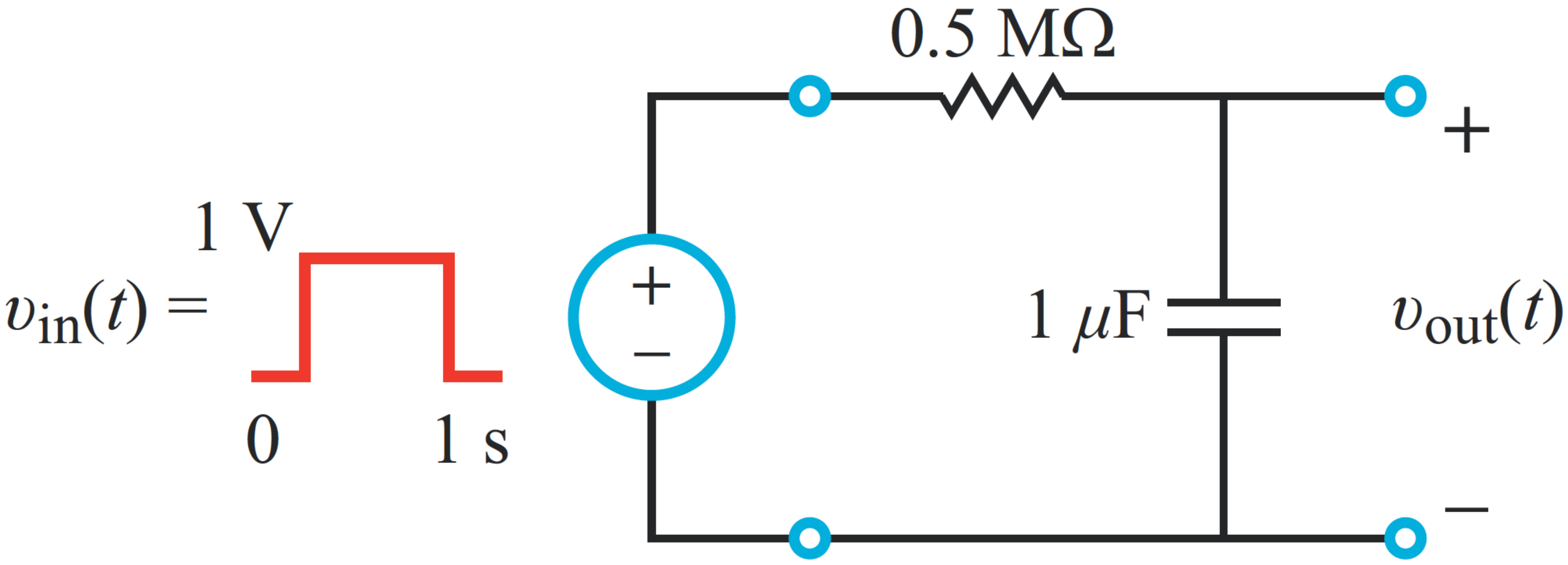


Practical note:

Focus on the points where the pattern changes, and fill in the gaps

Previous Example Revisited

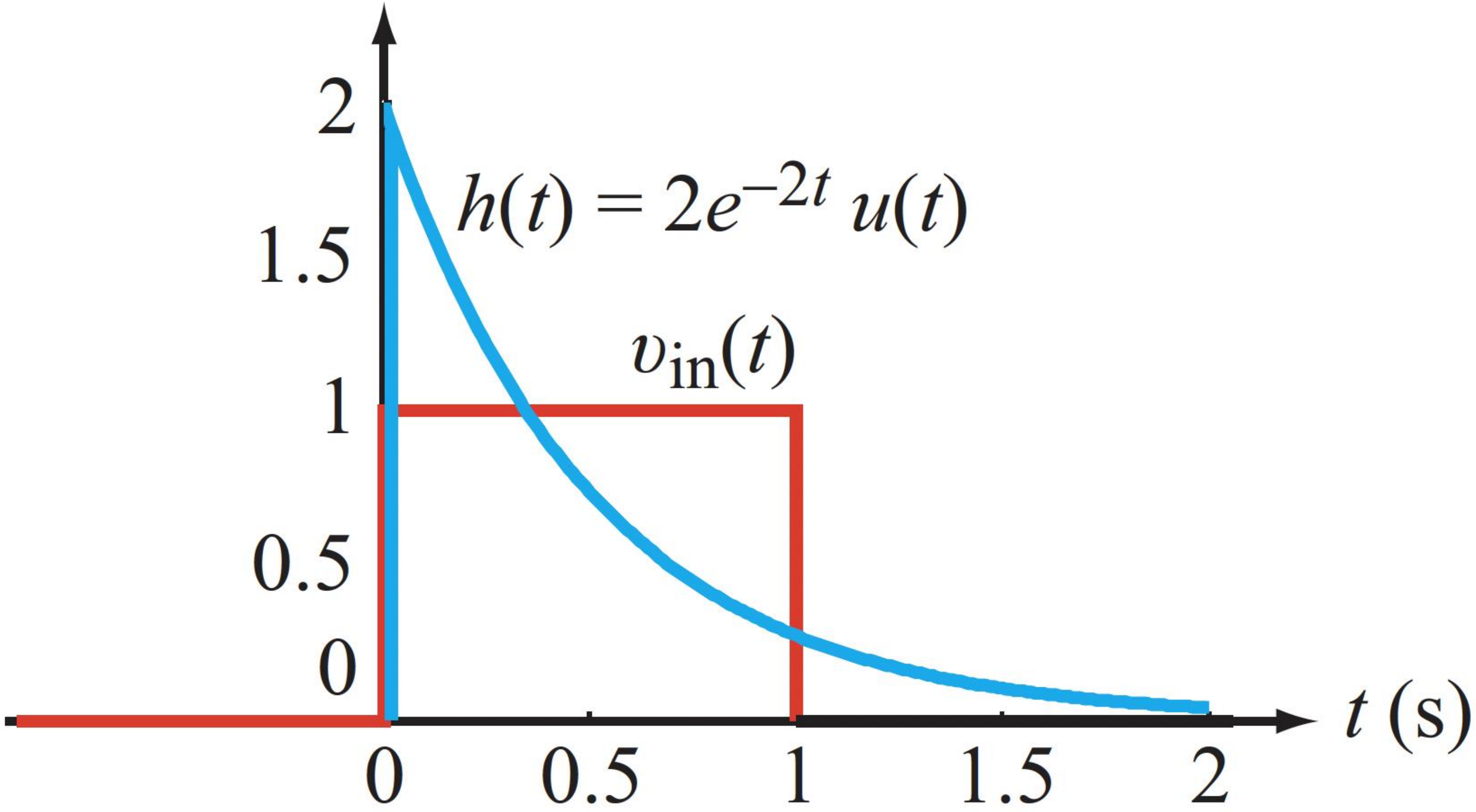
Let us apply a pulse to this RC circuit:



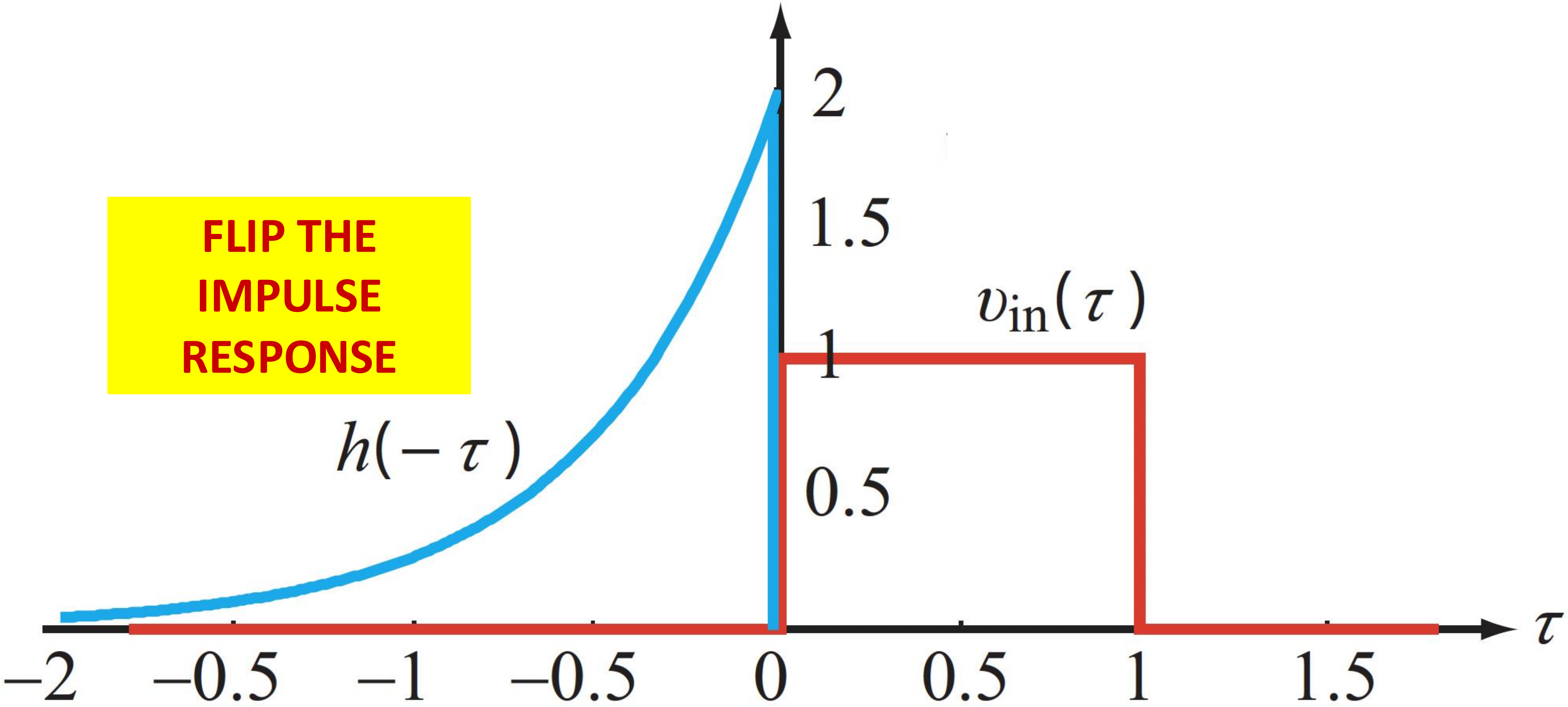
$$\tau_C = 0.5 \,\Box \quad h(t) = 2e^{-2t}u(t)$$

Convolution Problem

Let us start by sketching the signal and system:



Flip the Impulse Response

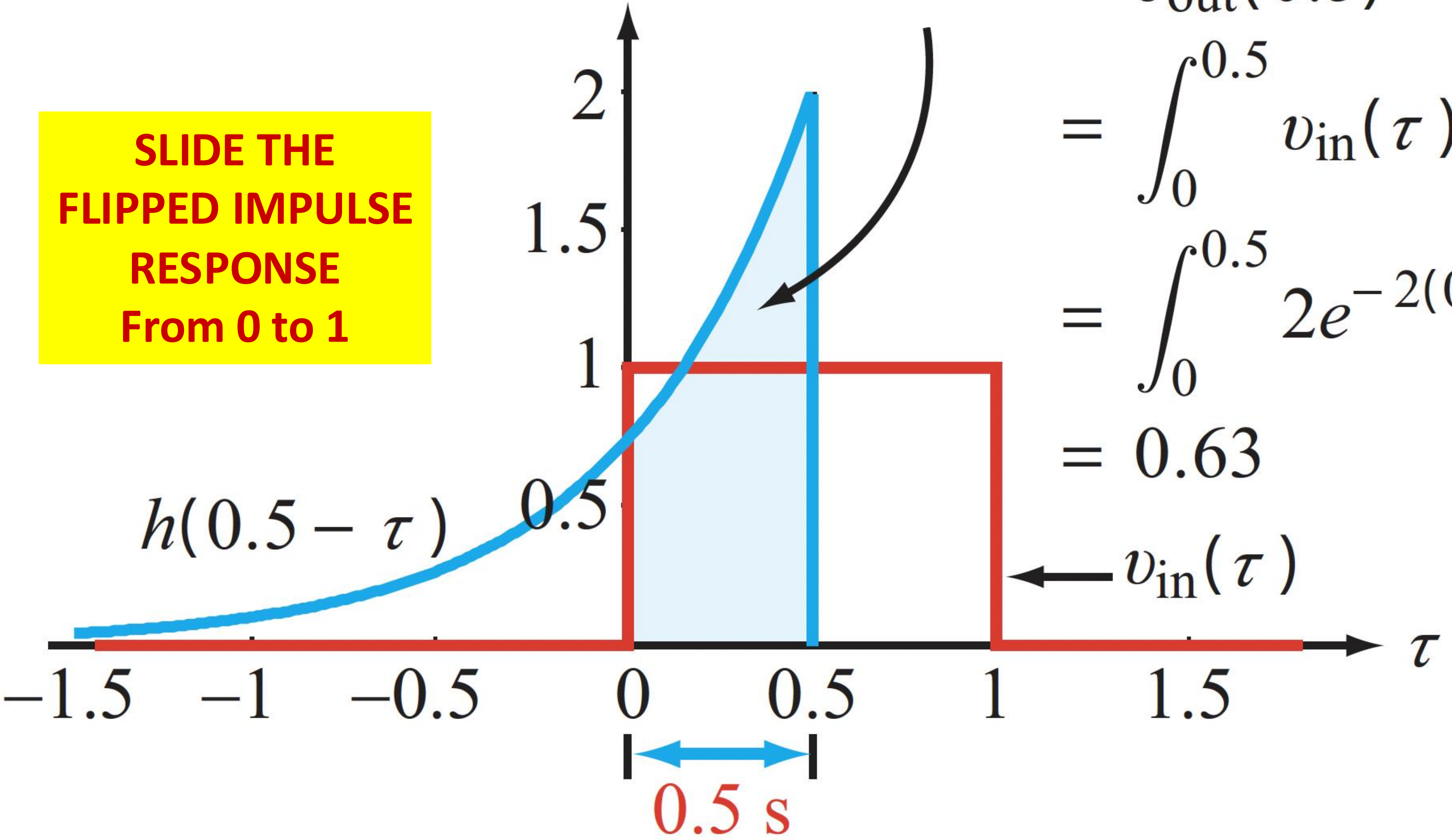


$$\text{From } -\square < t < 0, \text{ overlap} = 0$$

$$\square \quad y(t) = 0$$

Slide the Impulse Response

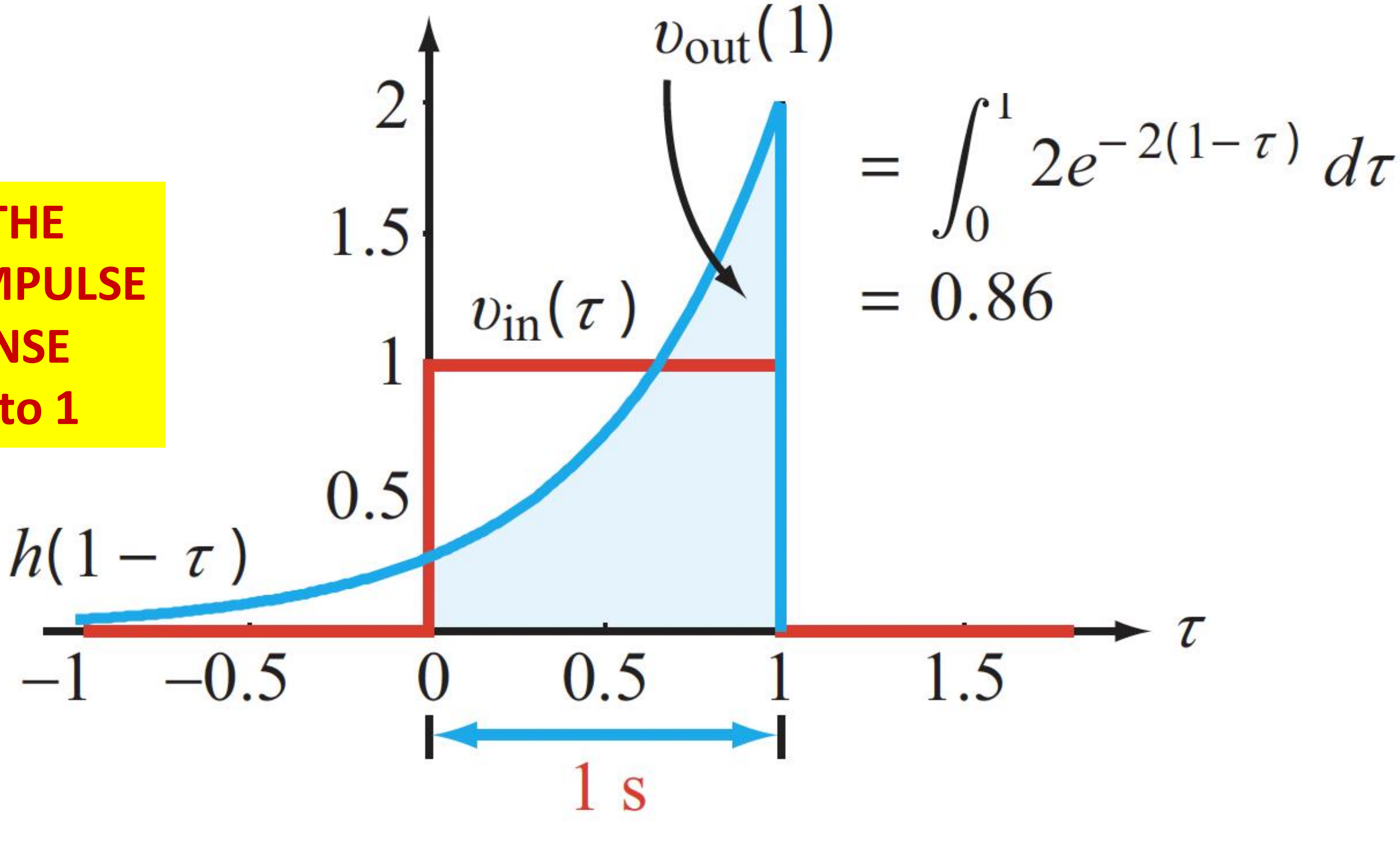
SLIDE THE FLIPPED IMPULSE RESPONSE From 0 to 1



$$v_{out}(0.5)$$
$$= \int_0^{0.5} v_{in}(\tau)\, h(0.5-\tau)\, d\tau$$
$$= \int_0^{0.5} 2e^{-2(0.5-\tau)}\, d\tau$$
$$= 0.63$$

Slide the Impulse Response

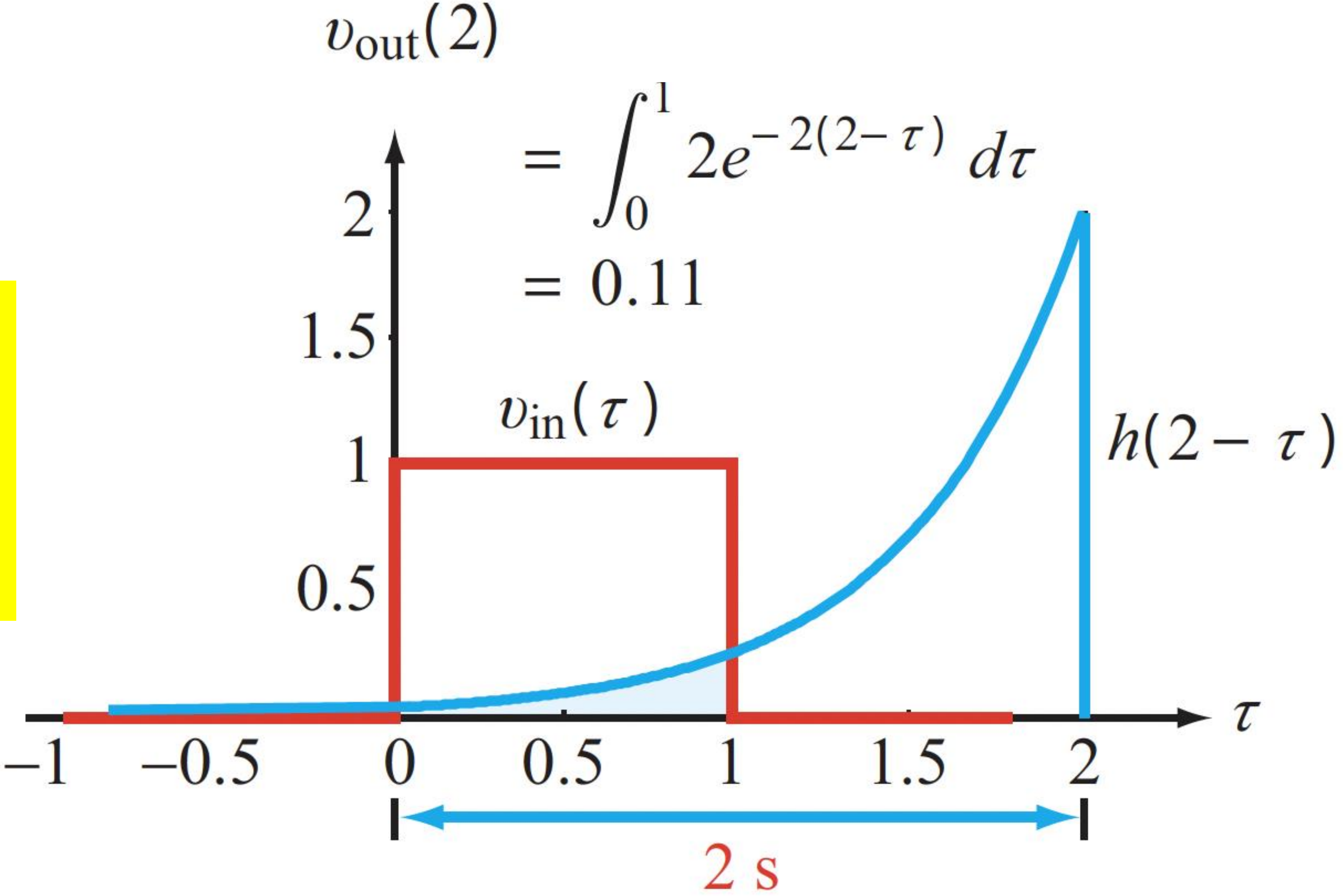
SLIDE THE FLIPPED IMPULSE RESPONSE From 0 to 1



(d) **Convolution @ $t = 1$ s**

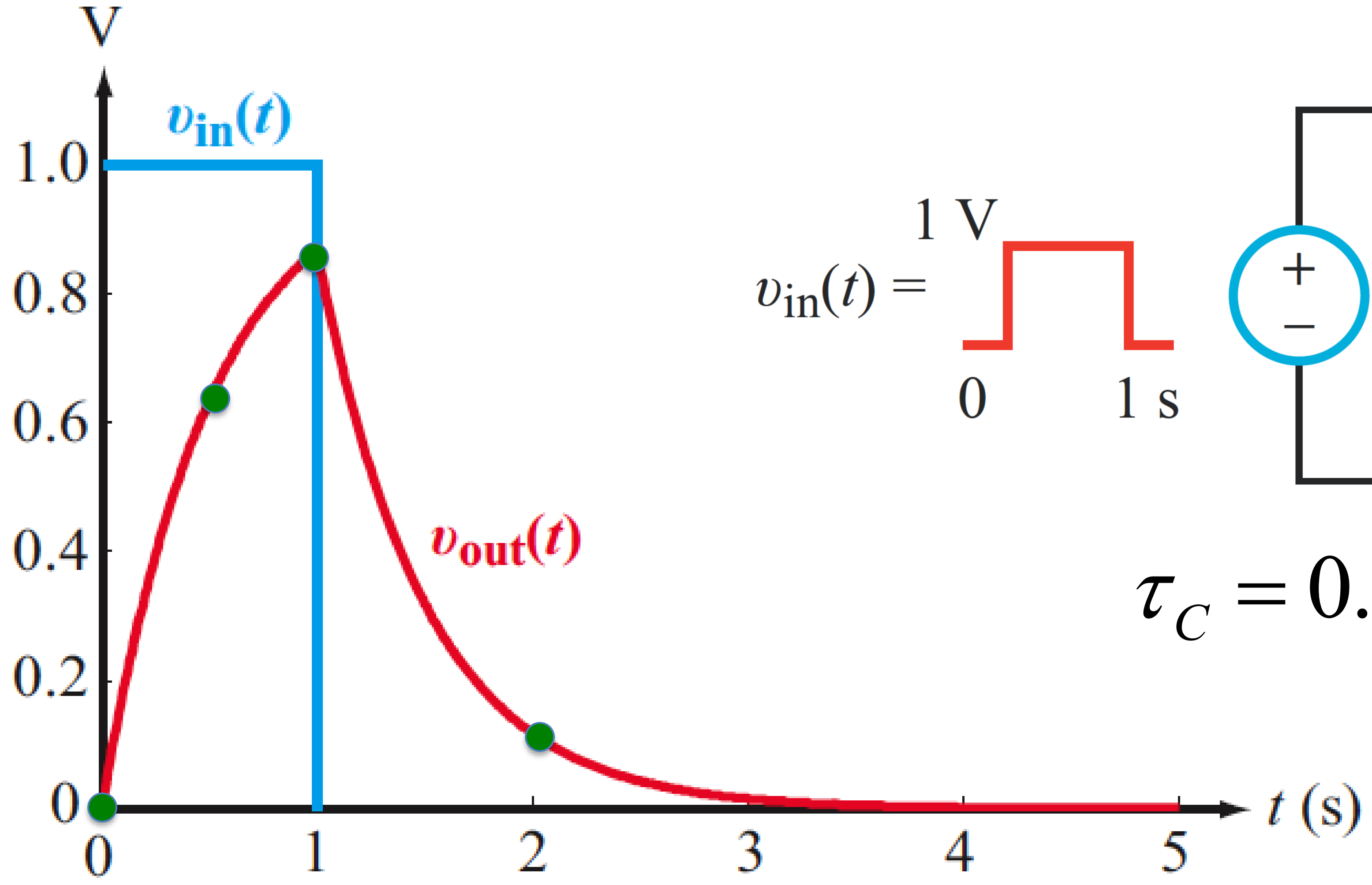
Slide the Impulse Response

SLIDE THE FLIPPED IMPULSE RESPONSE From 1 to 2 & beyond

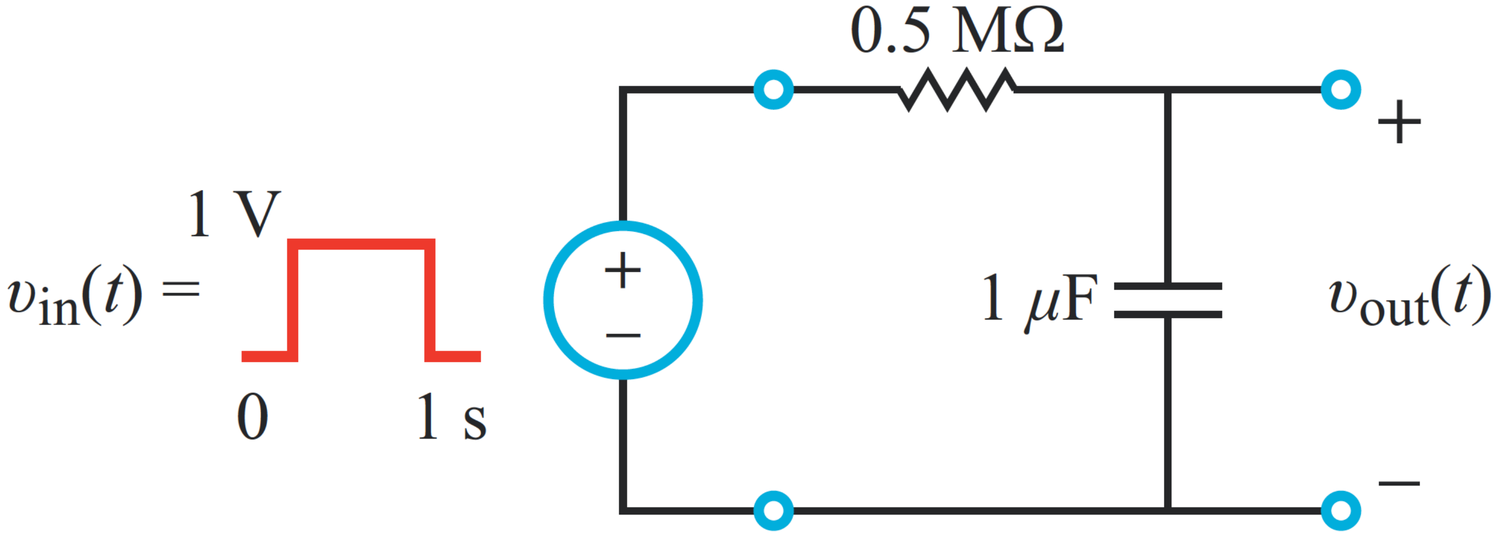


(e) **Convolution @ $t = 2$ s**

Assemble the Final Answer



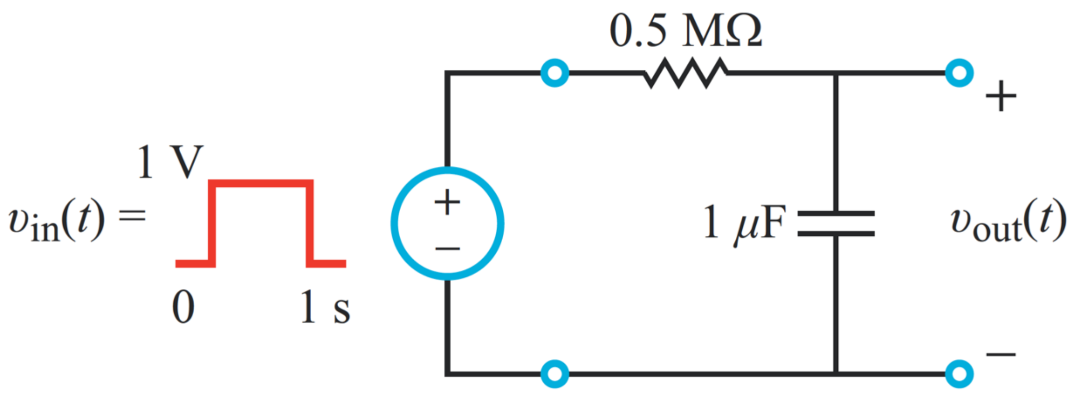
(b) Output response

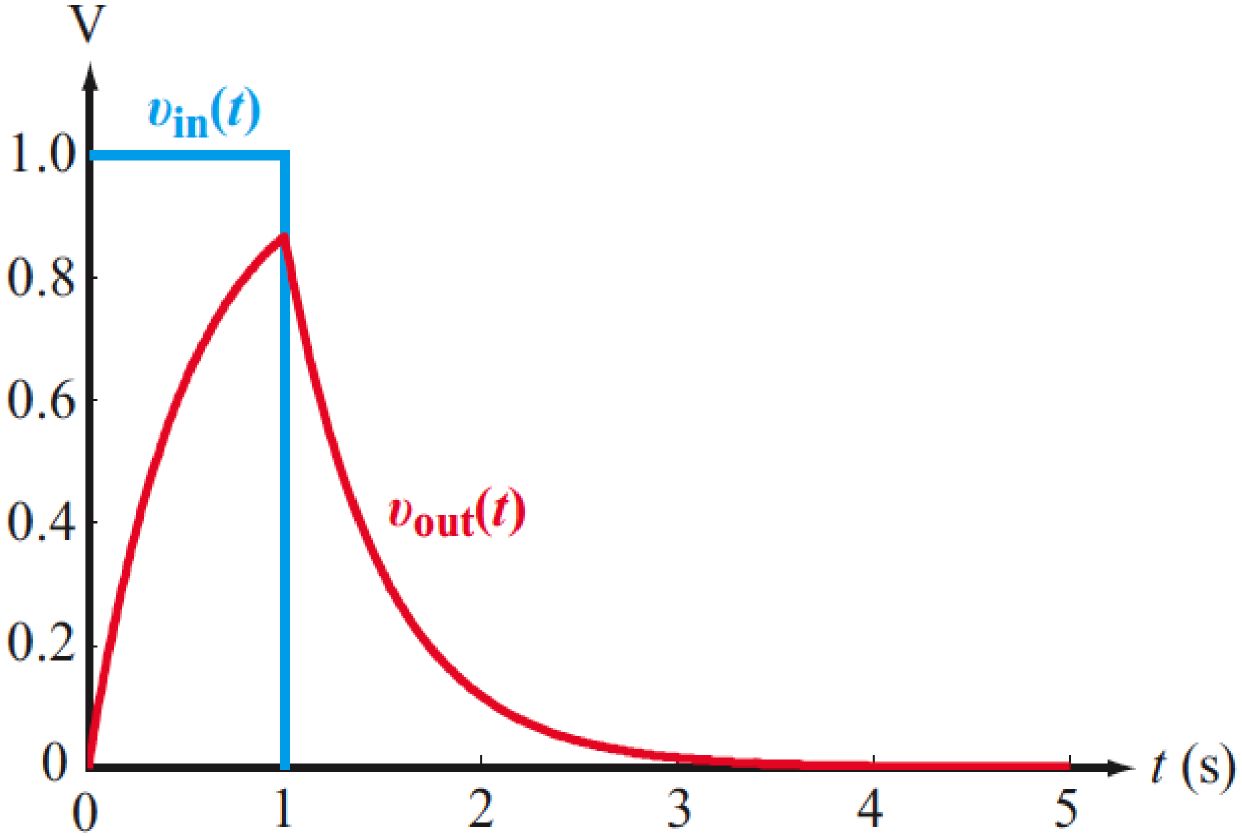


$$\tau_C = 0.5 \square \quad h(t) = 2e^{-2t}u(t)$$

Recall: RC Circuit

We did this problem before:

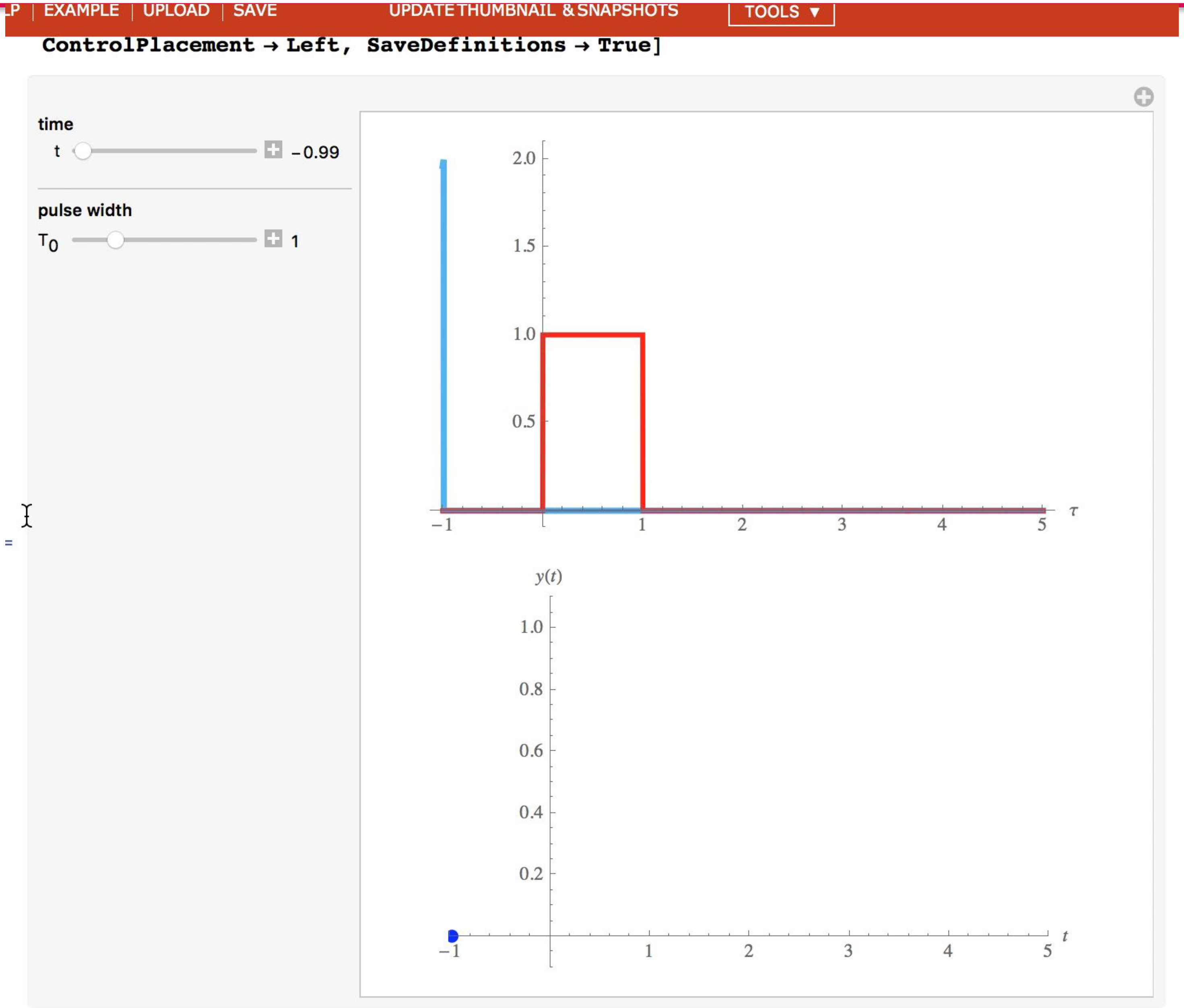




(b) Output response

$$\tau_C = 0.5\ \square \quad h(t) = 2e^{-2t}u(t)$$

Animation of the Convolution



Convolution Properties

Property #1: Commutative Property

$$y(t) = h(t) * x(t) = x(t) * h(t)$$

$$x(t) \rightarrow \boxed{h(t)} \rightarrow y(t)$$

$$h(t) \rightarrow \boxed{x(t)} \rightarrow y(t)$$

Practical value: We can swap the input and impulse response and the result is the same

Insight: Signals and systems are mathematically the same !

Proving Commutativity

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\,d\tau = x(t) * h(t)$$

$$\text{Change of variable: } \tau = t - u$$

$$\Rightarrow \quad \tau = t - u, \text{ and } d\tau = -du$$

$$y(t) = \int_{-\infty}^{+\infty} x(t-u)h(u)(-du) = \int_{\infty}^{-\infty} x(t-u)h(u)(-du)$$

$$= \int_{-\infty}^{+\infty} x(t-u)h(u)\,du = h(t) * x(t)$$

In summary, $x(t) * h(t) = h(t) * x(t)$

The Step Response

Test Signal $\qquad$ **Output**

$$u(t) \longrightarrow \boxed{\text{LTI System}} \longrightarrow y_{step}(t)$$

We can calculate the unit step response using the convolution integral:

$$y_{step}(t) = \int_{-\infty}^{+\infty} u(\tau)h(t-\tau)\,d\tau$$

$$= \int_{0}^{t} h(t-\tau)\,d\tau = \int_{0}^{t} h(\tau)\,d\tau \qquad \textbf{For Causal System}$$

The Step Response

$u(t) \rightarrow$ LTI System $\rightarrow y_{step}(t)$

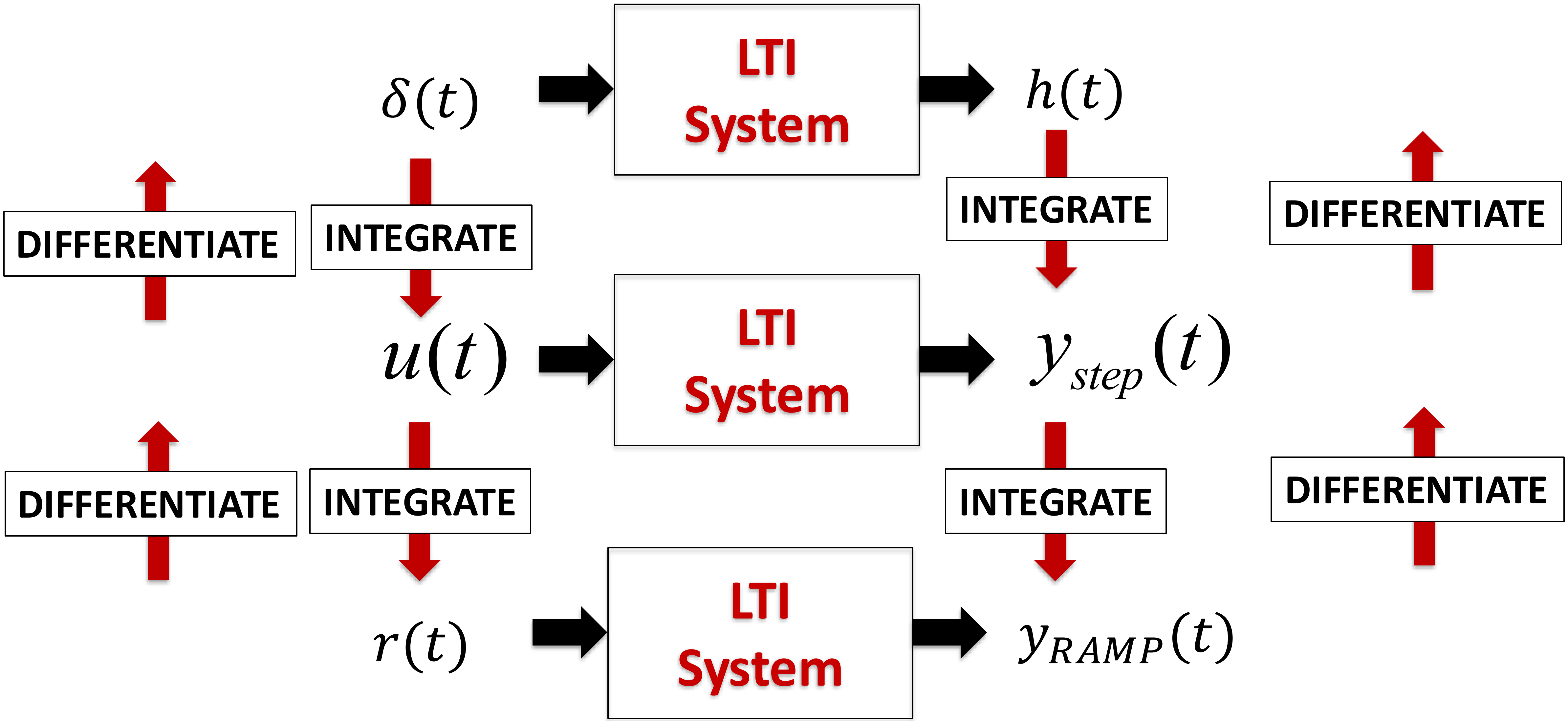
The unit step response is easy to produce experimentally with just a switch

- Unlike the impulse, no extreme timing or voltages are required
- The smoother output can be recorded conveniently and reliably

From the recorded unit step response, we can calculate the impulse response by differentiation:

$$h(t) = \frac{dy_{step}(t)}{dt}$$

Basic Patterns



Associative Property

$$x(t) \rightarrow \boxed{h_1(t)} \rightarrow z(t) \rightarrow \boxed{h_2(t)} \rightarrow y(t)$$

SERIES CONNECTION

$$z(t) = x(t) * h_1(t)$$

$$y(t) = z(t) * h_2(t)$$

$$y(t) = [x(t) * h_1(t)] * h_2(t)$$

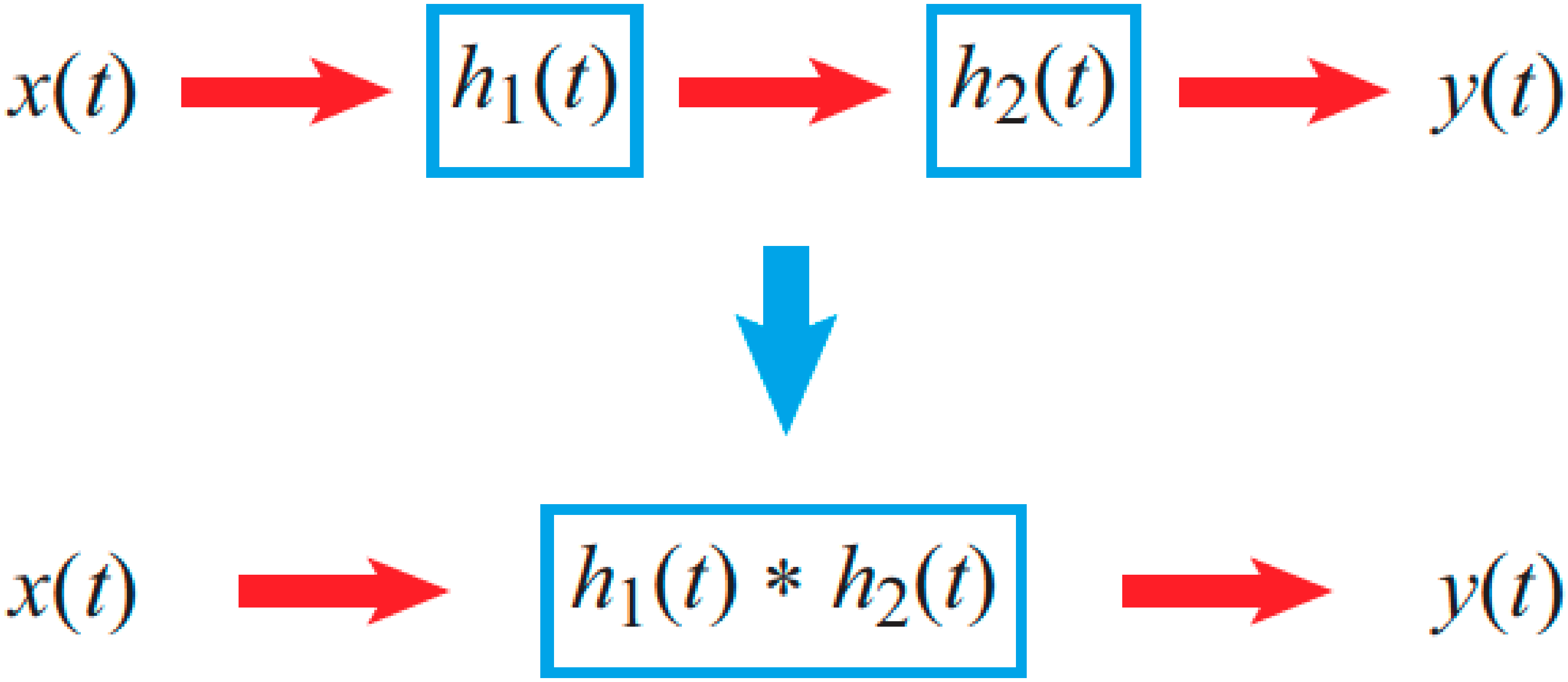
The associative property allows us to write this as:

$$y(t) = x(t) * [h_1(t) * h_2(t)]$$

The combined convolution of three functions is the same, regardless of the order in which it is carried out

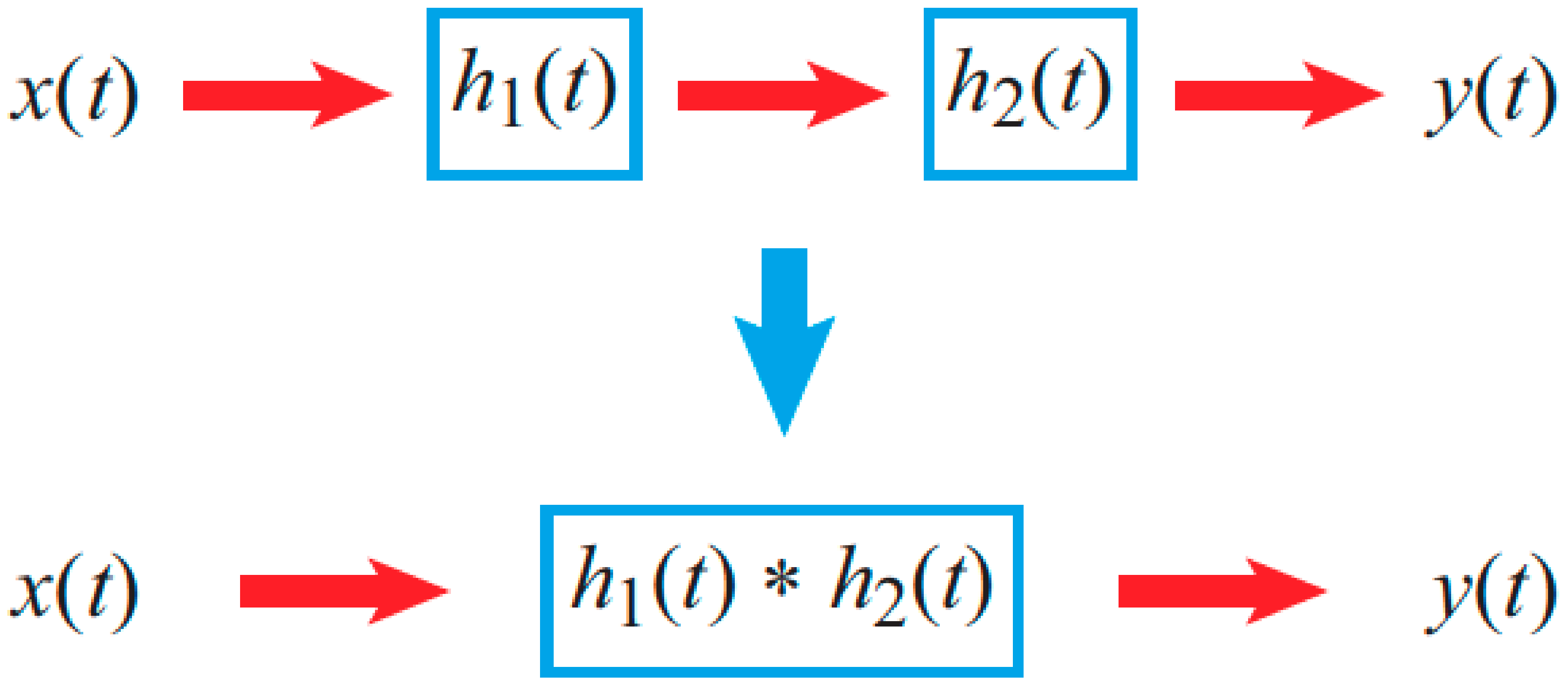
Equivalent Systems

<u>Practical Value</u>: When multiple LTI systems are cascaded in series, we can come up with a single "equivalent system" description:



EQUIVALENT SYSTEM

Useful Fact



If $h_1(t)$ and $h_2(t)$ are LTI systems, then the equivalent system is also LTI

Distributive Property

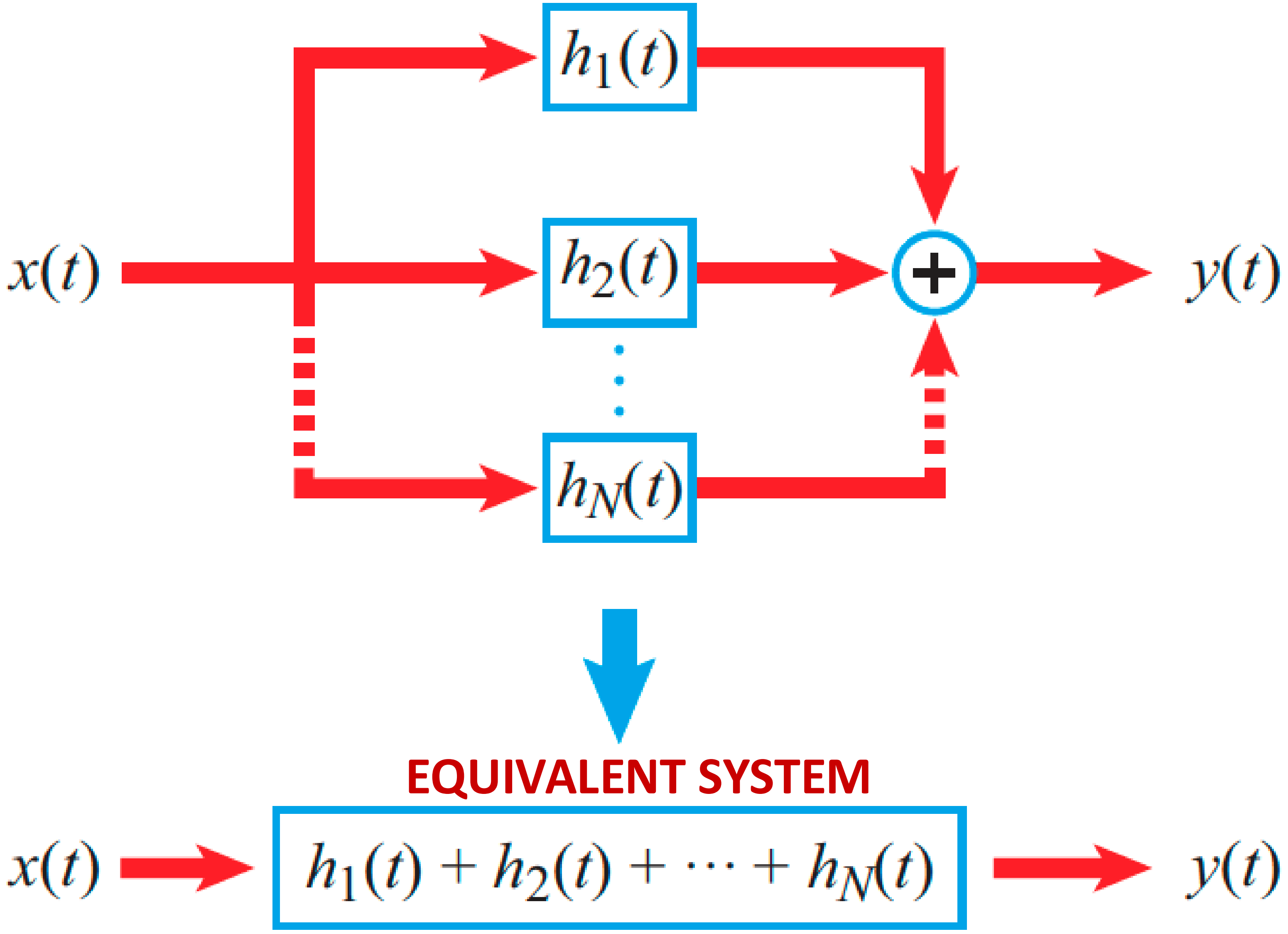
We can perform a convolution on the sum of signals:

$$x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$$

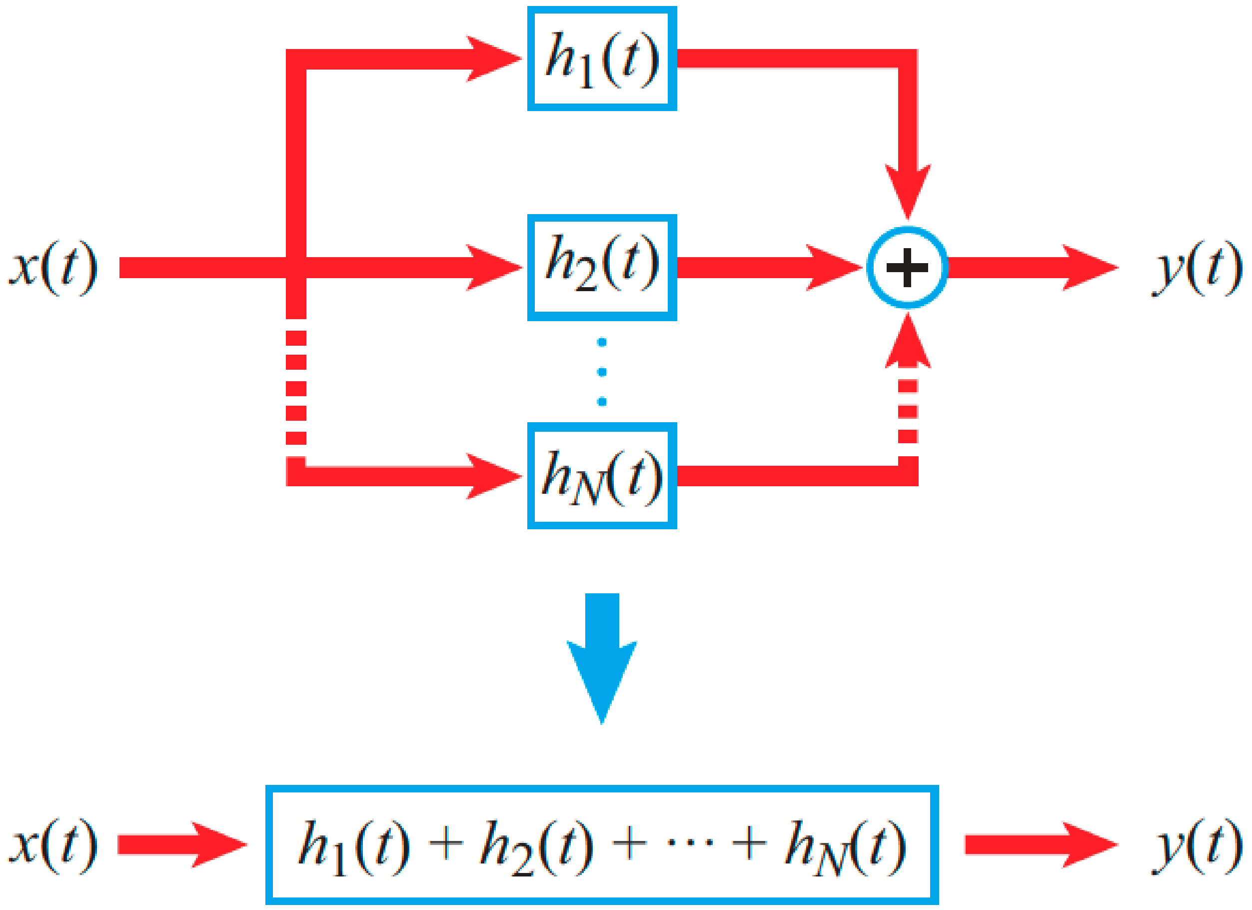
<u>Practical value</u>: We can break up a complicated system into a sum of simpler units, process them individually, and sum the results

$$h(t) = h_1(t) + h_2(t) + h_3(t) + \dots$$

Systems Connected in Parallel



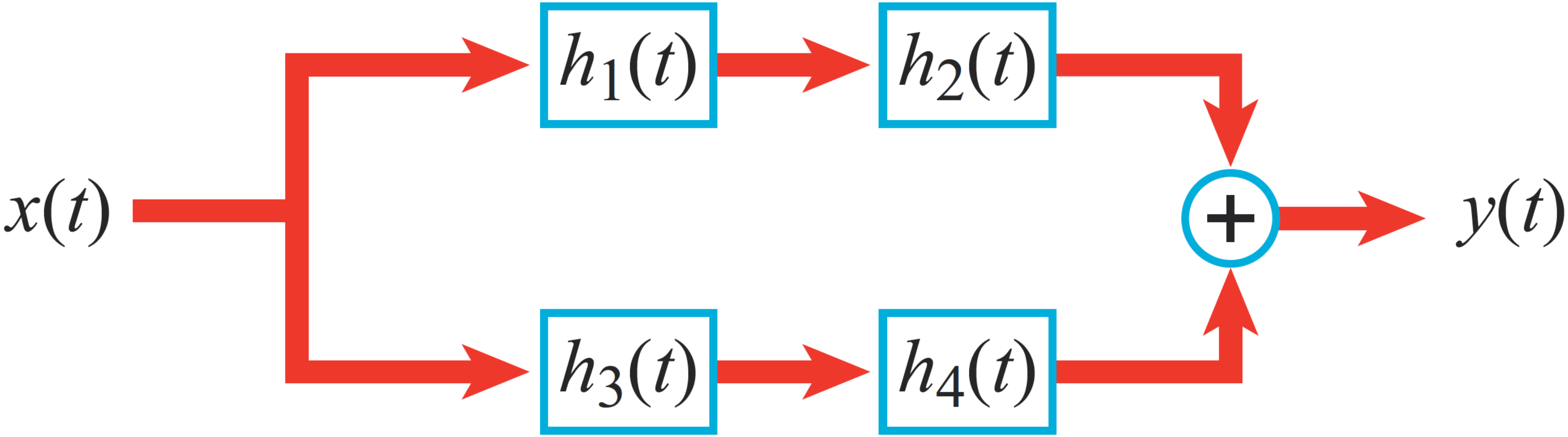
Interesting & Important Fact



If $h_1(t)$ and $h_2(t)$ are LTI systems, then the equivalent system is also LTI

Combined Example

Write the overall impulse response of the following system:



$$h(t) = h_1(t) * h_2(t) + h_3(t) * h_4(t)$$

IMPORTANT: THIS IS ALSO LTI

Time-shift Property

Delaying the signal or system delays the output by the total delay

$$x(t-T_1) = x(t) * \delta(t-T_1)$$

$$h(t-T_2) = h(t) * \delta(t-T_2)$$

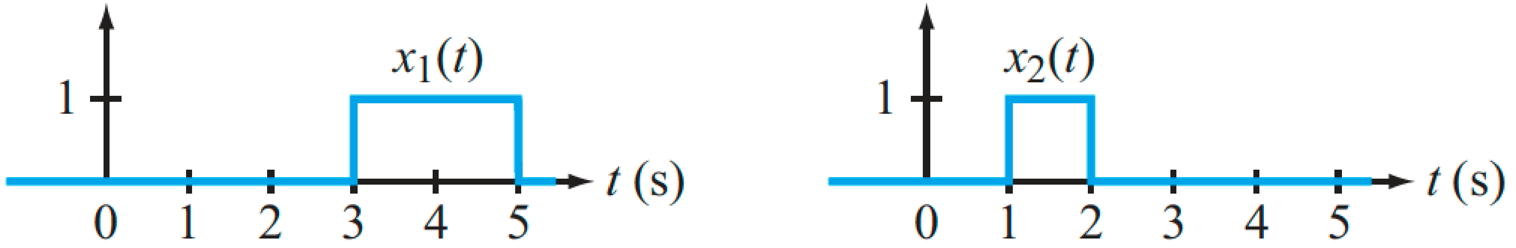
$$x(t-T_1) * h(t-T_2) = x(t) * h(t) * \delta(t-T_1) * \delta(t-T_2)$$

$$= y(t) * \delta(t-T_1-T_2)$$

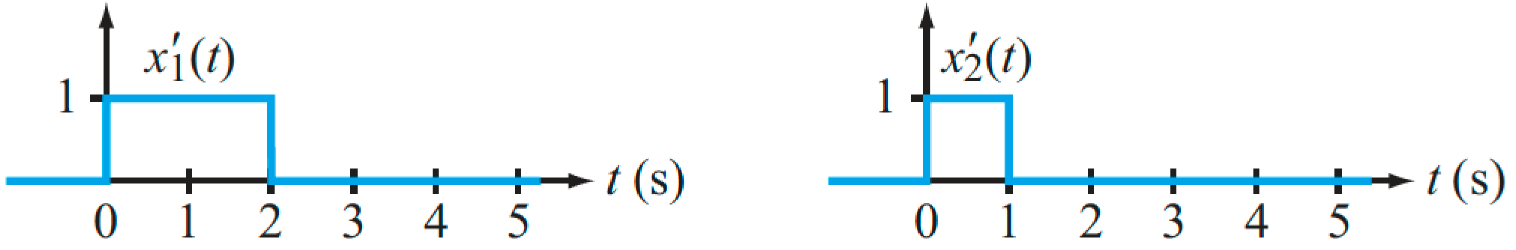
$$= y(t-T_1-T_2)$$ **The delays add up**

□ $x(t-T_1) * h(t-T_2) = y(t-T_1-T_2)$

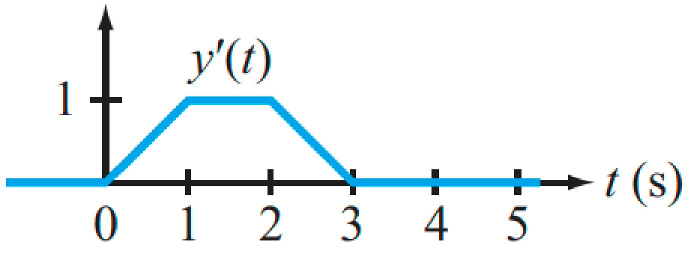
Example: Time-shift Property is Useful



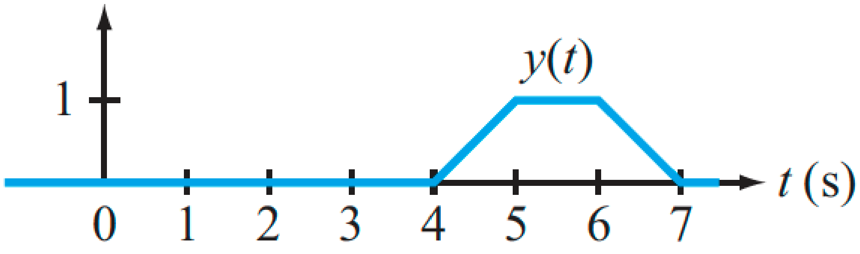
(a) **Original pulses**



(b) **Pulses shifted to start at $t = 0$**

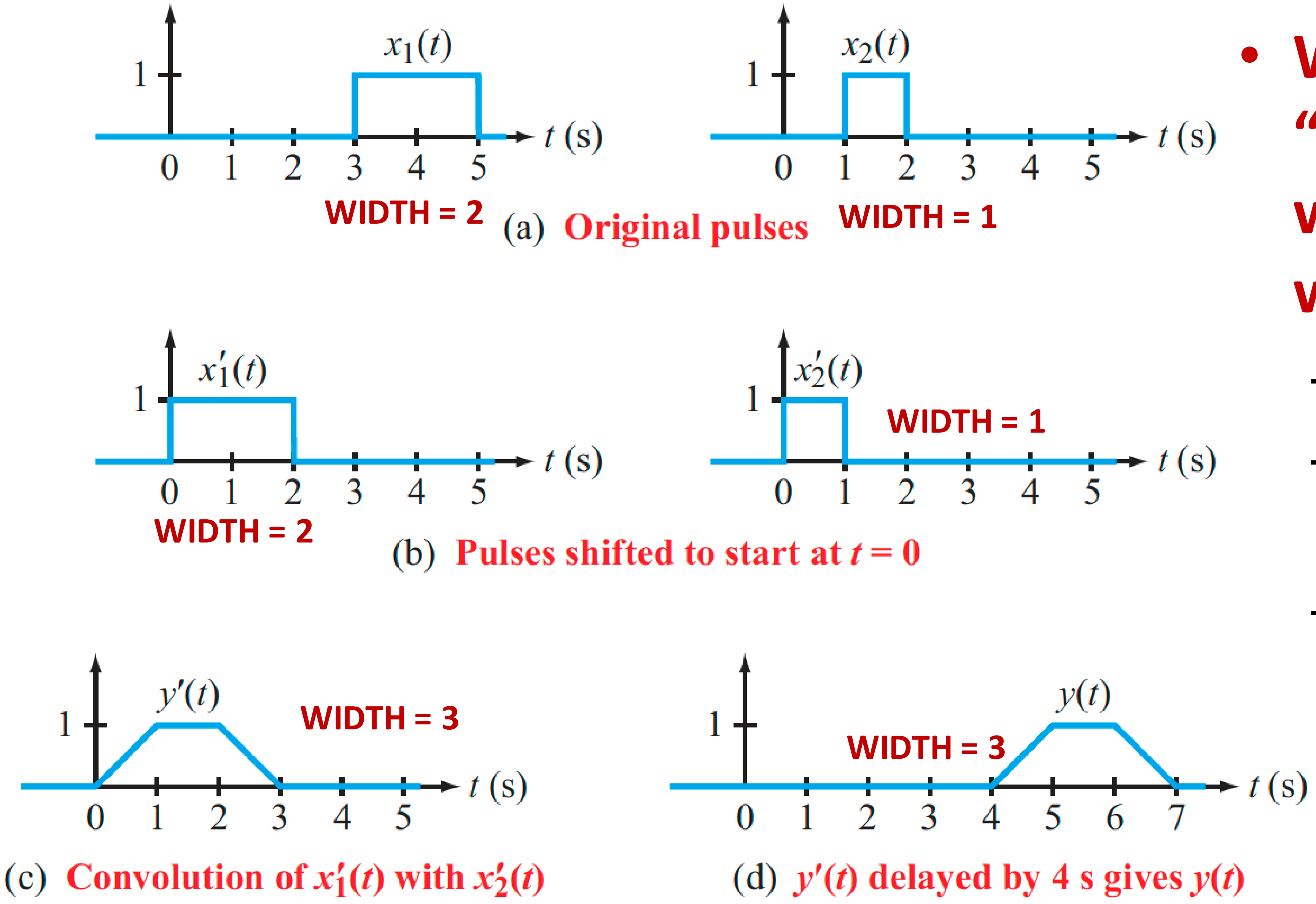


(c) **Convolution of $x_1'(t)$ with $x_2'(t)$**



(d) **$y'(t)$ delayed by 4 s gives $y(t)$**

Observe the Widths



- **We observe that the "width" of the output was the sum of the widths of the inputs**
  - This is true in general
  - Easy to understand graphically
  - "width" is the duration for which the signal is non-zero

Differentiation Property

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

$$\frac{dy(t)}{dt} = \frac{d}{dt}\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

The integration is with respect to τ

$$= \int_{-\infty}^{\infty} x(\tau)\frac{d}{dt}[h(t-\tau)]d\tau$$

$$\frac{dy(t)}{dt} = x(t) * \frac{d}{dt}[h(t)] = \frac{d}{dt}[x(t)] * h(t)$$

The Differentiator System

The differentiator is an LTI system that we can cascade in any order

$$\frac{dy(t)}{dt} = x(t) * \frac{d}{dt}[h(t)] = \frac{d}{dt}[x(t)] * h(t)$$

$$x(t) \longrightarrow \boxed{\frac{d}{dt}[\ ]} \longrightarrow \boxed{h(t)} \longrightarrow \frac{dy(t)}{dt}$$

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow \boxed{\frac{d}{dt}[\ ]} \longrightarrow \frac{dy(t)}{dt}$$

The Integrator System

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

Consider the special case when $h(t) = u(t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)u(t-\tau)\,d\tau = \int_{-\infty}^{t} x(\tau)\,d\tau$$

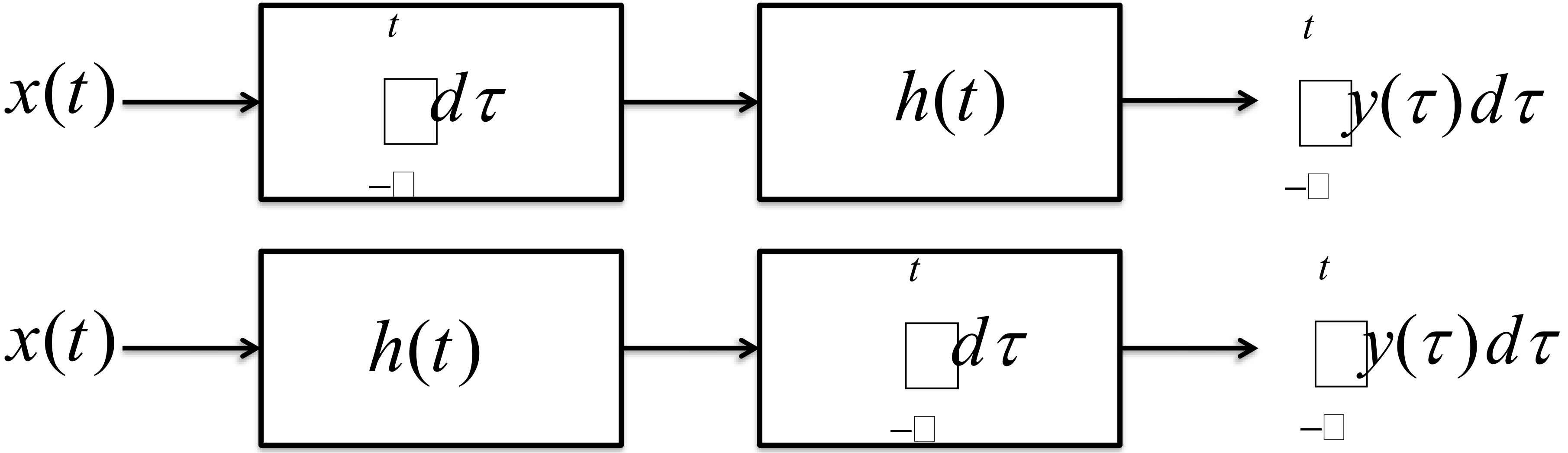
$$x(t) \longrightarrow \boxed{u(t)} \longrightarrow x(t) * u(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$$

$$x(t) \longrightarrow \boxed{\int_{-\infty}^{t} d\tau} \longrightarrow \int_{-\infty}^{t} x(\tau)\,d\tau$$

Running Integral

The Integrator

THE INTEGRATOR IS ALSO AN LTI SYSTEM THAT WE CAN CASCADE IN ANY ORDER



$$\int_{-\infty}^{t} y(\tau)d\tau = x(t) * \left[\int_{-\infty}^{t} h(\tau)d\tau\right] = \left[\int_{-\infty}^{t} x(\tau)d\tau\right] * h(t)$$

INTEGRATION PROPERTY

Summary of Basic Tricks

$$x(t) * \delta(t - T) = x(t - T)$$

DELAYER

$$x(t) * u(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$

UNIVERSAL INTEGRATOR

$$u(t) * u(t) = r(t)$$

INTEGRATE THE UNIT STEP

$$r(t) * u(t) = \frac{1}{2} t^2 u(t)$$

INTEGRATE THE RAMP

CONVOLUTIONS PROPERTIES CAN HELP SIMPLIFY OUR WORK, and SOMETIMES DO CONVOLUTIONS WITHOUT EXPLICIT INTEGRATION

Example

Compute the following convolution without computing any integrals:

$$y(t) = \big(u(t-1) + \delta(t-2)\big) * \big(u(t) - r(t-1)\big)$$

Hint: Use the convolution properties

Example

Compute the following convolution without computing any integrals:

$$y(t) = \big(u(t-1) + \delta(t-2)\big) * (u(t) - r(t-1))$$

Solution:

Expand it out

$$y(t) = u(t-1) * u(t) - u(t-1) * r(t-1) + \delta(t-2) * u(t) - \delta(t-2) * r(t-1)$$

Apply our basic tricks for each term, & add the delays

$$= r(t-1) - \frac{1}{2}(t-2)^2 u(t-2) + u(t-2) - r(t-3)$$

Summary of Convolution Properties

MEMORIZE

1. Commutative	$x(t) * h(t) = h(t) * x(t)$
2. Associative	$[g(t) * h(t)] * x(t) = g(t) * [h(t) * x(t)]$
3. Distributive	$x(t) * [h_1(t) + \cdots + h_N(t)] = x(t) * h_1(t) + \cdots + x(t) * h_N(t)$
4. Causal $*$ Causal $=$ Causal	$y(t) = u(t) \int\limits_0^t h(\tau)\, x(t-\tau)\, d\tau$
5. Time-shift	$h(t-T_1) * x(t-T_2) = y(t-T_1-T_2)$
6. Convolution with Impulse	$x(t) * \delta(t-T) = x(t-T)$
7. Width	Width of $y(t)$ = width of $x(t)$ + width of $h(t)$
8. Area	Area of $y(t)$ = area of $x(t)$ $\times$ area of $h(t)$
9. Convolution with $u(t)$	$y(t) = x(t) * u(t) = \int\limits_{-\infty}^t x(\tau)\, d\tau$ (Ideal integrator)
10a. Differentiation	$\left(\dfrac{d^m x}{dt^m}\right) * \left(\dfrac{d^n h}{dt^n}\right) = \dfrac{d^{m+n} y}{dt^{m+n}}$
10b. Integration	$\int\limits_{-\infty}^t y(\tau)\, d\tau = x(t) * \left[\int\limits_{-\infty}^t h(\tau)\, d\tau\right] = \left[\int\limits_{-\infty}^t x(\tau)\, d\tau\right] * h(t)$

The Next Class (#7) will be Different

You will be asked to hand-derive all the convolutions in Table 2-2

- Make sure to come fully prepared
  - Review integration, especially integration by parts
  - Watch our YouTube lectures as needed
  - Memorize the convolution integral. You should be able to write it down from memory, without making any mistakes
  - Review Lecture 5, especially the part where we figure out the limits of integration and range of validity for convolution integrals
  - Feel welcome to get an early start

I am here to help

- Ask questions and get answers
- I will walk around the class and help you if you get stuck.